Week 2 – Dataset Exploration and First Wrangling

Selecting, Cleaning, and Understanding Your Data

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1 Overview

Introduce the purpose of Week 2 and what this document covers.

- Key aims of this week
- Expected outcomes
- General encouragement statement

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2 Dataset Selection and Rationale

2.1 1.1 Dataset choice

Explain what dataset you are using and why.

We have chosen the data on Life expectancy from the World Health Organisation under the Topic: “Global Health Estimates: Life Expectancy and leading causes of death and disability”. We believe this will aid us in identifying discrepencies and Regional inequalities in both High and low income country’s Life expectancy. The data set also allows us to filter between life Expectancy at Birth and at 60. We chose the GDP data from the World Bank Group to help us classify the countries in the life expectancy data by their income levels eg high income, lower middle income and upper middle income and high income.

2.1.1 (Insert code cells – load dataset)

▶ Show code

IndicatorCode	Indicator	ValueType	ParentLocationCod
<chr>	<chr>	<chr>	<chr>
WHOSIS_000001	Life expectancy at birth (years)	text	AFR
WHOSIS_000001	Life expectancy at birth (years)	text	AFR
WHOSIS_000001	Life expectancy at birth (years)	text	AFR
WHOSIS_000001	Life expectancy at birth (years)	text	AFR
WHOSIS_000001	Life expectancy at birth (years)	text	EMR
WHOSIS_000001	Life expectancy at birth (years)	text	AFR
WHOSIS_000001	Life expectancy at birth (years)	text	EMR
WHOSIS_000001	Life expectancy at birth (years)	text	AFR
WHOSIS_000001	Life expectancy at birth (years)	text	AFR
WHOSIS_000001	Life expectancy at birth (years)	text	AFR

1-10 of 10,000 rows | 1-4 of 34 columns Previous **1** 2 3 4 5 6 … 1000 Next

▶ Show code

Country Name <chr>	Country Code <chr>
Aruba	ABW
Africa Eastern and Southern	AFE
Afghanistan	AFG
Africa Western and Central	AFW
Angola	AGO
Albania	ALB
Andorra	AND
Arab World	ARB
United Arab Emirates	ARE
Argentina	ARG

1-10 of 266 rows | 1-2 of 69 columns Previous **1** 2 3 4 5 6 … 27 Next

2.2 1.2 Justification

Describe why this dataset suits your project idea.

The data has access to 194 countries allowing us to provide a representative sample of the disparities. This allows us to filter the data by parameters that highlight regional inequality. This will serve as a basis for finding reasoning in any gaps identified. We will also be using GDP data closely with our original dataset in order to classify countries by income.

3 2 First Glimpse of the Data

3.1 2.1 Initial inspection

Discuss the structure, dimensions, and variable types.

The life expectancy dataset contains 24,420 rows and 34 columns, capturing detailed global health data by country, year, and sex. It includes mostly categorical variables like country, region, and sex, along with numeric variables such as life expectancy values and confidence ranges. This structure allows for demographic and regional comparisons over time. In contrast, the GDP dataset is smaller, with 266 rows and 69 columns, consisting mainly of numeric variables representing GDP per capita from 1960 onward for each country. Together, these datasets provide complementary socioeconomic and health insights that can be merged for analysis of how economic conditions relate to life expectancy across nations and years.

(Insert code cells – glimpse, summary)

▶ Show code

```
Rows: 24,420
Columns: 34
$ IndicatorCode              <chr> "WHOSIS_000001", "WHOSIS_000001", "WHOSIS_0…
$ Indicator                  <chr> "Life expectancy at birth (years)", "Life e…
$ ValueType                  <chr> "text", "text", "text", "text", "text", "te…
$ ParentLocationCode         <chr> "AFR", "AFR", "AFR", "AFR", "EMR", "AFR", "…
$ ParentLocation             <chr> "Africa", "Africa", "Africa", "Africa", "Ea…
$ `Location type`            <chr> "Country", "Country", "Country", "Country",…
$ SpatialDimValueCode        <chr> "LSO", "CAF", "LSO", "SWZ", "SOM", "CAF", "…
$ Location                   <chr> "Lesotho", "Central African Republic", "Les…
$ `Period type`              <chr> "Year", "Year", "Year", "Year", "Year", "Ye…
$ Period                     <dbl> 2021, 2021, 2021, 2021, 2021, 2021, 2021, 2…
$ IsLatestYear               <lgl> TRUE, TRUE, TRUE, TRUE, TRUE, TRUE, TRUE, T…
$ `Dim1 type`                <chr> "Sex", "Sex", "Sex", "Sex", "Sex", "Sex", "…
$ Dim1                       <chr> "Male", "Male", "Both sexes", "Male", "Male…
$ Dim1ValueCode              <chr> "SEX_MLE", "SEX_MLE", "SEX_BTSX", "SEX_MLE"…
$ `Dim2 type`                <lgl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,…
$ Dim2                       <lgl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,…
$ Dim2ValueCode              <lgl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,…
$ `Dim3 type`                <lgl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,…
$ Dim3                       <lgl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,…
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$ DataSourceDimValueCode     <lgl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,…
$ DataSource                 <lgl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,…
$ FactValueNumericPrefix     <lgl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,…
$ FactValueNumeric           <dbl> 48.73, 49.57, 51.48, 51.64, 51.75, 52.31, 5…
$ FactValueUoM               <lgl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,…
$ FactValueNumericLowPrefix  <lgl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,…
$ FactValueNumericLow        <dbl> 47.80, 48.45, 50.49, 50.73, 50.62, 51.06, 5…
$ FactValueNumericHighPrefix <lgl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,…
$ FactValueNumericHigh       <dbl> 49.70, 50.92, 52.57, 52.82, 53.10, 53.36, 5…
$ Value                      <chr> "48.7 [47.8-49.7]", "49.6 [48.5-50.9]", "51…
$ FactValueTranslationID     <lgl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,…
$ FactComments               <lgl> NA, NA, NA, NA, NA, NA, NA, NA, NA, NA, NA,…
$ Language                   <chr> "EN", "EN", "EN", "EN", "EN", "EN", "EN", "…
$ DateModified               <dttm> 2024-08-01 23:00:00, 2024-08-01 23:00:00, …
```

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 IndicatorCode       Indicator          ValueType         ParentLocationCode
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 Class :character   Class :character   Class :character   Class :character
 Mode  :character   Mode  :character   Mode  :character   Mode  :character



 ParentLocation     Location type      SpatialDimValueCode   Location
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 Class :character   Class :character   Class :character      Class :character
 Mode  :character   Mode  :character   Mode  :character      Mode  :character
```

```
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Mode  :character   Median :2010   TRUE :1110      Mode  :character
                   Mean   :2010
                   3rd Qu.:2016
                   Max.   :2021

    Dim1           Dim1ValueCode      Dim2 type        Dim2
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Class :character   Class :character   NA's:24420     NA's:24420
Mode  :character   Mode  :character



Dim2ValueCode  Dim3 type        Dim3         Dim3ValueCode
Mode:logical   Mode:logical   Mode:logical   Mode:logical
NA's:24420     NA's:24420     NA's:24420     NA's:24420



DataSourceDimValueCode DataSource     FactValueNumericPrefix FactValueNumeric
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NA's:24420             NA's:24420     NA's:24420             1st Qu.:18.77
                                                             Median :32.84
                                                             Mean   :44.59
                                                             3rd Qu.:71.87
                                                             Max.   :87.37

FactValueUoM   FactValueNumericLowPrefix FactValueNumericLow
Mode:logical   Mode:logical               Min.   : 8.67
NA's:24420     NA's:24420                 1st Qu.:18.44
                                          Median :32.52
                                          Mean   :44.21
                                          3rd Qu.:71.45
                                          Max.   :87.32
                                          NA's   :40
FactValueNumericHighPrefix FactValueNumericHigh    Value
Mode:logical               Min.   : 9.47        Length:24420
NA's:24420                 1st Qu.:19.14        Class :character
                           Median :33.23        Mode  :character
                           Mean   :45.01
                           3rd Qu.:72.29
                           Max.   :87.42
                           NA's   :40
FactValueTranslationID FactComments     Language
Mode:logical           Mode:logical   Length:24420
NA's:24420             NA's:24420     Class :character
                                      Mode  :character
```

```
  DateModified
 Min.   :2024-08-01 23:00:00
 1st Qu.:2024-08-01 23:00:00
 Median :2024-08-01 23:00:00
 Mean   :2024-08-01 23:00:00
 3rd Qu.:2024-08-01 23:00:00
 Max.   :2024-08-01 23:00:00
```

▶ Show code

```
Rows: 266
Columns: 69
$ `Country Name`   <chr> "Aruba", "Africa Eastern and Southern", "Afghanistan"…
$ `Country Code`   <chr> "ABW", "AFE", "AFG", "AFW", "AGO", "ALB", "AND", "ARB…
$ `Indicator Name` <chr> "GDP per capita (current US$)", "GDP per capita (curr…
$ `Indicator Code` <chr> "NY.GDP.PCAP.CD", "NY.GDP.PCAP.CD", "NY.GDP.PCAP.CD",…
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$ `1961`           <dbl> NA, 186.94178, NA, 127.45419, NA, NA, NA, 212.88966, …
$ `1962`           <dbl> NA, 197.40240, NA, 133.82704, NA, NA, NA, 210.80541, …
$ `1963`           <dbl> NA, 225.44049, NA, 139.00829, NA, NA, NA, 225.80063, …
$ `1964`           <dbl> NA, 208.99975, NA, 148.54938, NA, NA, NA, 243.89743, …
$ `1965`           <dbl> NA, 226.87651, NA, 155.56522, NA, NA, NA, 256.18014, …
$ `1966`           <dbl> NA, 240.95523, NA, 162.11077, NA, NA, NA, 263.95084, …
$ `1967`           <dbl> NA, 243.81732, NA, 144.94348, NA, NA, NA, 276.93297, …
$ `1968`           <dbl> NA, 257.19008, NA, 146.18917, NA, NA, NA, 307.53680, …
$ `1969`           <dbl> NA, 281.62928, NA, 161.92141, NA, NA, NA, 329.43142, …
$ `1970`           <dbl> NA, 276.78206, NA, 218.67629, NA, NA, 3935.41128, 357…
$ `1971`           <dbl> NA, 294.86615, NA, 195.86200, NA, NA, 4169.69537, 402…
$ `1972`           <dbl> NA, 311.51902, NA, 230.07079, NA, NA, 4940.29696, 470…
$ `1973`           <dbl> NA, 389.79697, NA, 280.64930, NA, NA, 6151.02717, 602…
$ `1974`           <dbl> NA, 463.5498, NA, 368.3019, NA, NA, 7139.5745, 1124.9…
$ `1975`           <dbl> NA, 479.1622, NA, 413.3714, NA, NA, 7925.4158, 1205.4…
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$ `1977`           <dbl> NA, 518.4506, NA, 490.8478, NA, NA, 8167.9229, 1630.3…
$ `1978`           <dbl> NA, 571.7203, NA, 523.7881, NA, NA, 9409.5085, 1728.2…
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$ `1980`           <dbl> NA, 773.4395, NA, 757.0756, 729.1120, 590.6077, 12474…
$ `1981`           <dbl> NA, 777.8331, NA, 1324.5466, 657.9826, 663.2942, 1046…
$ `1982`           <dbl> NA, 725.7281, NA, 1162.1307, 634.2215, 668.4545, 9697…
$ `1983`           <dbl> NA, 732.5887, NA, 872.8327, 636.8328, 661.5468, 8082.…
$ `1984`           <dbl> NA, 650.5635, NA, 737.6749, 650.4911, 639.4847, 7808.…
$ `1985`           <dbl> NA, 554.4391, NA, 753.7998, 772.4688, 639.8659, 7911.…
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$ `1987`           <dbl> 8244.0457, 665.1198, NA, 581.5238, 770.1011, 674.7934…
$ `1988`           <dbl> 10056.2614, 704.4662, NA, 560.3806, 807.4396, 652.774…
$ `1989`           <dbl> 11507.2172, 728.5493, NA, 509.3267, 907.7479, 697.995…
$ `1990`           <dbl> 12187.5364, 822.7939, NA, 590.3214, 965.8668, 617.230…
$ `1991`           <dbl> 13233.9905, 864.5639, NA, 603.8789, 881.9195, 336.587…
$ `1992`           <dbl> 13892.60514, 733.24386, NA, 563.31008, 668.70602, 200…
$ `1993`           <dbl> 14700.9598, 709.6593, NA, 570.8886, 449.7279, 367.279…
$ `1994`           <dbl> 16055.2878, 701.0416, NA, 577.5487, 334.9736, 586.416…
```

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$ `1995`           <dbl> 16548.7174, 766.8206, NA, 863.9506, 404.2948, 911.320…
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$ `1997`           <dbl> 17750.0096, 767.6842, NA, 1088.9968, 521.7029, 728.54…
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$ `2001`           <dbl> 20740.1326, 626.1691, 138.7068, 530.6532, 533.5862, 1…
$ `2002`           <dbl> 21307.2483, 625.9988, 178.9541, 616.7960, 882.1478, 1…
$ `2003`           <dbl> 21949.4860, 809.9681, 198.8711, 694.5009, 992.6990, 1…
$ `2004`           <dbl> 23700.6320, 981.7446, 221.7637, 840.2521, 1266.2109, …
$ `2005`           <dbl> 24171.8371, 1115.7140, 254.1842, 1001.6046, 1916.4684…
$ `2006`           <dbl> 24845.6585, 1220.0025, 274.2186, 1236.6281, 2617.0520…
$ `2007`           <dbl> 26736.3089, 1362.3107, 376.2232, 1407.6421, 3141.0460…
$ `2008`           <dbl> 28171.9094, 1421.1186, 381.7332, 1668.8411, 4103.0669…
$ `2009`           <dbl> 25134.7712, 1386.5902, 452.0537, 1454.7283, 3136.6455…
$ `2010`           <dbl> 24093.1401, 1601.7277, 560.6215, 1663.9669, 3597.3429…
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$ `2012`           <dbl> 25119.6655, 1702.9691, 651.4171, 1941.3696, 5086.0274…
$ `2013`           <dbl> 25813.5767, 1673.2211, 637.0871, 2131.6621, 5057.7479…
$ `2014`           <dbl> 26129.8391, 1656.1677, 625.0549, 2221.0107, 5005.9997…
$ `2015`           <dbl> 27458.2253, 1479.6153, 565.5697, 1860.7277, 3213.9026…
$ `2016`           <dbl> 27441.5297, 1329.8073, 522.0822, 1630.0394, 1807.9529…
$ `2017`           <dbl> 28440.0520, 1520.2122, 525.4698, 1574.2306, 2437.2597…
$ `2018`           <dbl> 30082.1276, 1538.9017, 491.3372, 1720.1403, 2538.5914…
$ `2019`           <dbl> 31096.2051, 1493.8179, 496.6025, 1798.3407, 2189.8557…
$ `2020`           <dbl> 22855.9323, 1344.1032, 510.7871, 1680.0393, 1449.9229…
$ `2021`           <dbl> 27200.0611, 1522.3933, 356.4962, 1765.9548, 1925.8747…
$ `2022`           <dbl> 30559.5335, 1628.3189, 357.2612, 1796.6686, 2929.6945…
$ `2023`           <dbl> 33984.7906, 1510.7430, 413.7579, 1599.3930, 2309.5341…
$ `2024`           <dbl> NA, 1567.6358, NA, 1284.1544, 2122.0837, 10011.6280, …
```

▶ Show code

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 Country Name       Country Code       Indicator Name     Indicator Code
 Length:266         Length:266         Length:266         Length:266
 Class :character   Class :character   Class :character   Class :character
 Mode  :character   Mode  :character   Mode  :character   Mode  :character
```

```
      1960              1961              1962              1963
 Min.   :  25.08   Min.   :  27.27   Min.   :  27.95   Min.   :  25.8
 1st Qu.: 123.30   1st Qu.: 127.45   1st Qu.: 129.53   1st Qu.: 138.9
 Median : 220.07   Median : 205.02   Median : 208.30   Median : 228.9
 Mean   : 470.95   Mean   : 490.32   Mean   : 537.85   Mean   : 570.7
 3rd Qu.: 499.87   3rd Qu.: 538.36   3rd Qu.: 574.69   3rd Qu.: 610.2
 Max.   :3002.09   Max.   :3062.97   Max.   :4748.39   Max.   :4444.5
 NA's   :115       NA's   :112       NA's   :110       NA's   :110
      1964              1965              1966              1967
 Min.   :  17.33   Min.   :  15.12   Min.   :  11.8   Min.   :  16.55
 1st Qu.: 144.58   1st Qu.: 155.79   1st Qu.: 164.1   1st Qu.: 161.73
 Median : 241.38   Median : 270.40   Median : 278.6   Median : 281.53
 Mean   : 616.90   Mean   : 671.95   Mean   : 715.7   Mean   : 729.06
 3rd Qu.: 707.38   3rd Qu.: 851.13   3rd Qu.: 895.8   3rd Qu.: 844.84
```

```
Max.   :4378.78   Max.   :4023.99   Max.   :4176.1   Max.   :4331.58
NA's   :110       NA's   :104       NA's   :103      NA's   :99
     1968              1969              1970              1971
Min.   :  21.55   Min.   :  21.51   Min.   :   20.72   Min.   :   21.13
1st Qu.: 166.23   1st Qu.: 175.15   1st Qu.:  200.49   1st Qu.:  208.78
Median : 300.66   Median : 315.05   Median :  352.82   Median :  383.55
Mean   : 762.98   Mean   : 826.62   Mean   :  995.67   Mean   : 1095.33
3rd Qu.: 847.35   3rd Qu.: 901.09   3rd Qu.: 1294.89   3rd Qu.: 1366.85
Max.   :4691.63   Max.   :5025.09   Max.   :12098.20   Max.   :13334.40
NA's   :98        NA's   :98        NA's   :76         NA's   :75
     1972              1973              1974              1975
Min.   :   23.29   Min.   :   24.8   Min.   :   41.41   Min.   :   35.16
1st Qu.:  228.30   1st Qu.:  279.5   1st Qu.:  338.35   1st Qu.:  364.25
Median :  427.22   Median :  538.0   Median :  701.08   Median :  717.76
Mean   : 1270.76   Mean   : 1617.1   Mean   : 2158.56   Mean   : 2325.07
3rd Qu.: 1558.48   3rd Qu.: 2171.9   3rd Qu.: 2560.81   3rd Qu.: 2715.12
Max.   :16187.43   Max.   :20882.0   Max.   :23822.97   Max.   :27920.74
NA's   :75         NA's   :75        NA's   :75         NA's   :71
     1976              1977              1978              1979
Min.   :   39.17   Min.   :   27.86   Min.   :   29.26   Min.   :   29.21
1st Qu.:  398.76   1st Qu.:  441.79   1st Qu.:  502.70   1st Qu.:  557.44
Median :  762.78   Median :  862.68   Median :  982.60   Median : 1113.62
Mean   : 2532.71   Mean   : 2778.29   Mean   : 3146.15   Mean   : 3732.49
3rd Qu.: 2933.02   3rd Qu.: 3207.80   3rd Qu.: 3359.57   3rd Qu.: 4122.07
Max.   :31196.98   Max.   :35020.36   Max.   :37607.79   Max.   :44989.96
NA's   :71         NA's   :68         NA's   :69         NA's   :68
     1980              1981              1982              1983
Min.   :   31.25   Min.   :   32.81   Min.   :   42.95   Min.   :   39.29
1st Qu.:  606.06   1st Qu.:  652.41   1st Qu.:  626.53   1st Qu.:  621.07
Median : 1243.27   Median : 1290.99   Median : 1310.02   Median : 1363.44
Mean   : 4249.54   Mean   : 4127.07   Mean   : 3942.27   Mean   : 3778.47
3rd Qu.: 4609.16   3rd Qu.: 4947.16   3rd Qu.: 4598.76   3rd Qu.: 4475.30
Max.   :58005.34   Max.   :48793.44   Max.   :45074.55   Max.   :39075.84
NA's   :58         NA's   :58         NA's   :57         NA's   :57
     1984              1985              1986              1987
Min.   :   36.33   Min.   :   40.38   Min.   :   42.42   Min.   :   41.15
1st Qu.:  580.84   1st Qu.:  571.42   1st Qu.:  541.14   1st Qu.:  586.35
Median : 1319.23   Median : 1220.84   Median : 1425.16   Median : 1381.70
Mean   : 3717.65   Mean   : 3687.48   Mean   : 4202.65   Mean   : 4752.64
3rd Qu.: 4507.28   3rd Qu.: 4656.25   3rd Qu.: 5181.88   3rd Qu.: 5259.09
Max.   :36582.37   Max.   :37689.86   Max.   :52082.45   Max.   :62456.07
NA's   :55         NA's   :53         NA's   :53         NA's   :46
     1988              1989              1990              1991
Min.   :   39.92   Min.   :   51.34   Min.   :   53.12   Min.   :   22.95
1st Qu.:  570.51   1st Qu.:  586.04   1st Qu.:  650.59   1st Qu.:  624.39
Median : 1303.78   Median : 1356.28   Median : 1557.09   Median : 1482.64
Mean   : 5129.12   Mean   : 5254.63   Mean   : 5815.00   Mean   : 5840.52
3rd Qu.: 5679.99   3rd Qu.: 5928.95   3rd Qu.: 6529.21   3rd Qu.: 6824.34
Max.   :67180.58   Max.   :66815.71   Max.   :81734.87   Max.   :80997.25
NA's   :43         NA's   :43         NA's   :26         NA's   :25
     1992              1993              1994              1995
Min.   :   30.25   Min.   :   53.74   Min.   :   61.49   Min.   :   62.14
1st Qu.:  575.65   1st Qu.:  560.45   1st Qu.:  580.80   1st Qu.:  682.85
Median : 1447.85   Median : 1397.27   Median : 1493.45   Median : 1775.49
```

```
Mean   : 6123.31   Mean   : 5930.76   Mean   : 6282.89   Mean   : 7051.76
3rd Qu.: 6730.68   3rd Qu.: 6704.86   3rd Qu.: 6997.47   3rd Qu.: 7109.34
Max.   :88614.93   Max.   :82685.46   Max.   :86703.81   Max.   :99032.24
NA's   :24         NA's   :23         NA's   :22         NA's   :19
     1996               1997               1998               1999
Min.   :   71.41   Min.   :  108.0   Min.   :  126.5   Min.   :   96.32
1st Qu.:  721.20   1st Qu.:  728.5   1st Qu.:  687.8   1st Qu.:  615.94
Median : 1918.38   Median : 1898.4   Median : 1864.1   Median : 1812.58
Mean   : 7267.42   Mean   : 7242.0   Mean   : 7393.4   Mean   : 7557.66
3rd Qu.: 7333.77   3rd Qu.: 7944.6   3rd Qu.: 8169.1   3rd Qu.: 7862.97
Max.   :98561.01   Max.   :88581.7   Max.   :90947.2   Max.   :89698.20
NA's   :19         NA's   :17        NA's   :16        NA's   :16
     2000               2001               2002               2003
Min.   :  122.3   Min.   :  118.5   Min.   :  109.6   Min.   :   111.4
1st Qu.:  635.4   1st Qu.:  637.2   1st Qu.:  683.9   1st Qu.:   764.6
Median : 1817.6   Median : 1881.0   Median : 2092.4   Median :  2347.5
Mean   : 7559.4   Mean   : 7567.7   Mean   : 8171.0   Mean   :  9391.4
3rd Qu.: 8377.7   3rd Qu.: 9612.0   3rd Qu.:10586.1   3rd Qu.: 11487.2
Max.   :81789.0   Max.   :82402.8   Max.   :90051.4   Max.   :111110.5
NA's   :14        NA's   :13        NA's   :9         NA's   :9
     2004               2005               2006               2007
Min.   :   125.2   Min.   :   147.2   Min.   :   161.9   Min.   :   166.2
1st Qu.:   922.9   1st Qu.:  1067.9   1st Qu.:  1227.0   1st Qu.:  1379.8
Median :  2817.0   Median :  3146.8   Median :  3639.8   Median :  4396.3
Mean   : 10677.8   Mean   : 11572.8   Mean   : 12932.7   Mean   : 14710.0
3rd Qu.: 12886.4   3rd Qu.: 14368.6   3rd Qu.: 16279.4   3rd Qu.: 18154.8
Max.   :125160.2   Max.   :130538.7   Max.   :143083.5   Max.   :184558.6
NA's   :9          NA's   :9          NA's   :8          NA's   :8
     2008               2009               2010               2011
Min.   :   189.5   Min.   :   199.1   Min.   :   216.7   Min.   :   230.1
1st Qu.:  1555.8   1st Qu.:  1461.4   1st Qu.:  1664.0   1st Qu.:  1925.8
Median :  5023.6   Median :  4799.8   Median :  5418.2   Median :  6010.4
Mean   : 15776.3   Mean   : 14301.7   Mean   : 15028.0   Mean   : 16532.1
3rd Qu.: 19105.0   3rd Qu.: 17718.4   3rd Qu.: 19847.2   3rd Qu.: 20554.5
Max.   :204263.8   Max.   :169150.3   Max.   :161853.9   Max.   :179364.0
NA's   :7          NA's   :5          NA's   :5          NA's   :4
     2012               2013               2014               2015
Min.   :   231.7   Min.   :   234.8   Min.   :   250.5   Min.   :   281
1st Qu.:  1930.4   1st Qu.:  2079.8   1st Qu.:  2164.9   1st Qu.:  1997
Median :  6474.2   Median :  6706.9   Median :  6886.7   Median :  6141
Mean   : 16415.2   Mean   : 16990.7   Mean   : 17316.2   Mean   : 15671
3rd Qu.: 20093.6   3rd Qu.: 20521.6   3rd Qu.: 20638.8   3rd Qu.: 19061
Max.   :165444.7   Max.   :184940.7   Max.   :195693.6   Max.   :170437
NA's   :6          NA's   :6          NA's   :5          NA's   :7
     2016               2017               2018               2019
Min.   :   235.3   Min.   :   236.7   Min.   :   224.9   Min.   :   210.2
1st Qu.:  1952.0   1st Qu.:  2153.3   1st Qu.:  2147.0   1st Qu.:  2198.3
Median :  5968.8   Median :  6323.2   Median :  6765.5   Median :  6761.5
Mean   : 15793.0   Mean   : 16602.2   Mean   : 17601.8   Mean   : 17561.4
3rd Qu.: 19035.4   3rd Qu.: 20845.4   3rd Qu.: 22657.4   3rd Qu.: 22903.2
Max.   :173604.8   Max.   :170663.4   Max.   :188298.3   Max.   :193746.8
NA's   :8          NA's   :8          NA's   :8          NA's   :8
     2020               2021               2022               2023
Min.   :   210   Min.   :   214.1   Min.   :   250.6   Min.   :   192.1
```

```
 1st Qu.:  2082   1st Qu.:  2445.2   1st Qu.:  2388.6   1st Qu.:  2531.8
 Median :  6237   Median :  7051.2   Median :  7630.9   Median :  7856.9
 Mean   : 16317   Mean   : 18841.7   Mean   : 19495.4   Mean   : 19933.2
 3rd Qu.: 19966   3rd Qu.: 22132.4   3rd Qu.: 23994.4   3rd Qu.: 23114.7
 Max.   :176892   Max.   :223823.4   Max.   :226052.0   Max.   :256580.5
 NA's   :9        NA's   :9          NA's   :10         NA's   :18
      2024
 Min.   :   153.9
 1st Qu.:  2592.9
 Median :  7695.2
 Mean   : 18153.2
 3rd Qu.: 23919.0
 Max.   :138935.0
 NA's   :35
```

3.2 2.2 Observations

Summarise key insights about shape, columns, and data types.

The life expectancy dataset contains 34 columns with a mix of categorical and numeric data, such as country, region, gender, and numeric life expectancy values. It's well-organized for demographic analysis and comparison across countries and years. The GDP dataset is wider, with 69 columns and each year represented as a numeric column showing GDP per capita. This format captures long-term economic trends. Later in the project, we'll clean and merge these datasets by country and year to explore the relationship between economic status (GDP) and health outcomes (life expectancy).

4 3 Initial Data Wrangling

Document cleaning, transformations, and reasoning.

4.1 3.1 Cleaning

Describe how you handled missing values, duplicates, formats.

During data cleaning, we focused on restructuring, simplifying, and preparing both datasets for merging and analysis. For the GDP dataset, we transposed the year columns into a single "Year" column to make the data tidy and easier to work with over time. We then selected only the relevant columns; country name, code, year, and GDP per capita so as to remove unnecessary information. For the life expectancy data, we dropped columns that contained no useful or unique information and kept only those relevant to our analysis, such as location, period, gender (Dim1), and the numeric life expectancy value. We also renamed the "SpatialDimValueCode" column to "Country_Code" to ensure consistency with the GDP dataset for merging. The categorical variables like Indicator and Country_Code were converted to factors for better grouping and analysis. While inspecting the datasets, we noticed a number of missing (NA) values,particularly in the GDP data, but decided to keep them for now and address them later in the analysis phase through imputation, filtering, or exclusion where appropriate. No duplicates were found after the cleaning process.

(Insert code cells – cleaning steps)

▶ Show code

Country Name	Country Code	Year	GDP_per_capita
<chr>	<chr>	<dbl>	<dbl>
Aruba	ABW	1960	*NA*
Aruba	ABW	1961	*NA*
Aruba	ABW	1962	*NA*
Aruba	ABW	1963	*NA*
Aruba	ABW	1964	*NA*
Aruba	ABW	1965	*NA*
Aruba	ABW	1966	*NA*
Aruba	ABW	1967	*NA*
Aruba	ABW	1968	*NA*
Aruba	ABW	1969	*NA*

1-10 of 10,000 rows Previous **1** 2 3 4 5 6 … 1000 Next

▶ Show code

```
          IndicatorCode                  Indicator
                      0                          0
              ValueType         ParentLocationCode
                      0                          0
         ParentLocation             Location type
                      0                          0
     SpatialDimValueCode                   Location
                      0                          0
            Period type                     Period
                      0                          0
           IsLatestYear                  Dim1 type
                      0                          0
                   Dim1              Dim1ValueCode
                      0                          0
           Dim2 type                       Dim2
                  24420                      24420
        Dim2ValueCode                  Dim3 type
                  24420                      24420
                   Dim3              Dim3ValueCode
                  24420                      24420
DataSourceDimValueCode                 DataSource
                  24420                      24420
FactValueNumericPrefix           FactValueNumeric
                  24420                          0
          FactValueUoM  FactValueNumericLowPrefix
                  24420                      24420
   FactValueNumericLow FactValueNumericHighPrefix
                     40                      24420
  FactValueNumericHigh                      Value
                     40                          0
FactValueTranslationID                FactComments
                  24420                      24420
              Language                DateModified
                      0                          0
```

▶ Show code

Indicator	ParentLocation
<chr>	<chr>
Life expectancy at birth (years)	Africa
Life expectancy at birth (years)	Africa
Life expectancy at birth (years)	Africa
Life expectancy at birth (years)	Africa
Life expectancy at birth (years)	Eastern Mediterranean
Life expectancy at birth (years)	Africa
Life expectancy at birth (years)	Eastern Mediterranean
Life expectancy at birth (years)	Africa
Life expectancy at birth (years)	Africa
Life expectancy at birth (years)	Africa

1-10 of 10,000 rows | 1-2 of 7 columns Previous **1** 2 3 4 5 6 … 1000 Next

▶ Show code

Indicator	ParentLocation
<fct>	<fct>
Life expectancy at birth (years)	Africa
Life expectancy at birth (years)	Africa
Life expectancy at birth (years)	Africa
Life expectancy at birth (years)	Africa
Life expectancy at birth (years)	Eastern Mediterranean
Life expectancy at birth (years)	Africa
Life expectancy at birth (years)	Eastern Mediterranean
Life expectancy at birth (years)	Africa
Life expectancy at birth (years)	Africa
Life expectancy at birth (years)	Africa

1-10 of 10,000 rows | 1-2 of 7 columns Previous **1** 2 3 4 5 6 … 1000 Next

▶ Show code

```
[1] "Life expectancy at age 60 (years)" "Life expectancy at birth (years)"
```

4.2 3.2 Transformation, integration, validation

Conceptual or implemented; note decisions or possible future steps.

(Insert code cells – optional transformation)

▶ Show code

Location	Country Code	Year	GDP_per_capita
<chr>	<chr>	<dbl>	<dbl>
Aruba	ABW	2001	20740.1326

Location	Country Code	Year	GDP_per_capita
<chr>	<chr>	<dbl>	<dbl>
Aruba	ABW	2002	21307.2483
Aruba	ABW	2003	21949.4860
Aruba	ABW	2004	23700.6320
Aruba	ABW	2005	24171.8371
Aruba	ABW	2006	24845.6585
Aruba	ABW	2007	26736.3089
Aruba	ABW	2008	28171.9094
Aruba	ABW	2009	25134.7712
Aruba	ABW	2010	24093.1401

1-10 of 5,586 rows Previous **1** 2 3 4 5 6 … 559 Next

▶ Show code

Location	Country_Code	Period	GDP_per_capita
<chr>	<chr>	<dbl>	<dbl>
Aruba	ABW	2001	20740.1326
Aruba	ABW	2002	21307.2483
Aruba	ABW	2003	21949.4860
Aruba	ABW	2004	23700.6320
Aruba	ABW	2005	24171.8371
Aruba	ABW	2006	24845.6585
Aruba	ABW	2007	26736.3089
Aruba	ABW	2008	28171.9094
Aruba	ABW	2009	25134.7712
Aruba	ABW	2010	24093.1401

1-10 of 5,586 rows Previous **1** 2 3 4 5 6 … 559 Next

▶ Show code

Location	Country_Code	Period	GDP_per_capita
<fct>	<fct>	<dbl>	<dbl>
Aruba	ABW	2001	20740.1326
Aruba	ABW	2002	21307.2483
Aruba	ABW	2003	21949.4860
Aruba	ABW	2004	23700.6320
Aruba	ABW	2005	24171.8371
Aruba	ABW	2006	24845.6585
Aruba	ABW	2007	26736.3089
Aruba	ABW	2008	28171.9094
Aruba	ABW	2009	25134.7712
Aruba	ABW	2010	24093.1401

1-10 of 5,586 rows Previous **1** 2 3 4 5 6 … 559 Next

▶ Show code

Indicator	ParentLocation
<fct>	<fct>
Life expectancy at birth (years)	Africa
Life expectancy at birth (years)	Africa
Life expectancy at birth (years)	Africa
Life expectancy at birth (years)	Africa
Life expectancy at birth (years)	Eastern Mediterranean
Life expectancy at birth (years)	Africa
Life expectancy at birth (years)	Eastern Mediterranean
Life expectancy at birth (years)	Africa
Life expectancy at birth (years)	Africa
Life expectancy at birth (years)	Africa

1-10 of 10,000 rows | 1-2 of 7 columns Previous **1** 2 3 4 5 6 … 1000 Next

▶ Show code

Indicator	ParentLocation
<fct>	<fct>
Life expectancy at birth (years)	Africa
Life expectancy at birth (years)	Africa
Life expectancy at birth (years)	Africa
Life expectancy at birth (years)	Africa
Life expectancy at birth (years)	Eastern Mediterranean
Life expectancy at birth (years)	Africa
Life expectancy at birth (years)	Eastern Mediterranean
Life expectancy at birth (years)	Africa
Life expectancy at birth (years)	Africa
Life expectancy at birth (years)	Africa

1-10 of 10,000 rows | 1-2 of 7 columns Previous **1** 2 3 4 5 6 … 1000 Next

▶ Show code

Indicator	ParentLocation
<fct>	<fct>
Life expectancy at age 60 (years)	Africa
Life expectancy at age 60 (years)	Africa
Life expectancy at age 60 (years)	Africa
Life expectancy at age 60 (years)	Eastern Mediterranean
Life expectancy at age 60 (years)	Africa
Life expectancy at age 60 (years)	Americas
Life expectancy at age 60 (years)	Africa
Life expectancy at age 60 (years)	Africa
Life expectancy at age 60 (years)	Africa
Life expectancy at age 60 (years)	Eastern Mediterranean

1-10 of 10,000 rows | 1-2 of 10 columns Previous **1** 2 3 4 5 6 … 1000 Next

▶ Show code

Indicator	ParentLocation	Period	Gender
<fct>	<fct>	<dbl>	<fct>
Life expectancy at age 60 (years)	Africa	2021	Male
Life expectancy at age 60 (years)	Africa	2021	Male
Life expectancy at age 60 (years)	Africa	2021	Male
Life expectancy at age 60 (years)	Eastern Mediterranean	2021	Male
Life expectancy at age 60 (years)	Africa	2021	Male
Life expectancy at age 60 (years)	Americas	2021	Male
Life expectancy at age 60 (years)	Africa	2021	Both sexes
Life expectancy at age 60 (years)	Africa	2021	Both sexes
Life expectancy at age 60 (years)	Africa	2021	Male
Life expectancy at age 60 (years)	Eastern Mediterranean	2021	Both sexes

1-10 of 10,000 rows | 1-4 of 8 columns Previous **1** 2 3 4 5 6 … 1000 Next

▶ Show code

Indicator	ParentLocation	Period	Gender
<fct>	<fct>	<dbl>	<fct>
Life expectancy at birth (years)	Africa	2021	Male
Life expectancy at birth (years)	Africa	2021	Male
Life expectancy at birth (years)	Africa	2021	Both sexes
Life expectancy at birth (years)	Africa	2021	Male
Life expectancy at birth (years)	Eastern Mediterranean	2021	Male
Life expectancy at birth (years)	Africa	2021	Both sexes
Life expectancy at birth (years)	Eastern Mediterranean	2021	Both sexes
Life expectancy at birth (years)	Africa	2021	Female
Life expectancy at birth (years)	Africa	2021	Both sexes
Life expectancy at birth (years)	Africa	2021	Male

1-10 of 10,000 rows | 1-4 of 9 columns Previous **1** 2 3 4 5 6 … 1000 Next

▶ Show code

```
                          Indicator                   ParentLocation
Life expectancy at age 60 (years):    0   Africa                :3102
Life expectancy at birth (years) :12210   Americas              :2244
                                          Eastern Mediterranean:1452
                                          Europe                :3300
                                          South-East Asia       : 726
                                          Western Pacific       :1386

    Period            Gender     Avg_Life_Expectancy Country_Code
Min.   :2000   Both sexes:4070   Min.   :36.60       Length:12210
1st Qu.:2005   Female    :4070   1st Qu.:64.19       Class :character
Median :2010   Male      :4070   Median :71.87       Mode  :character
```

```
Mean   :2010                    Mean   :70.17
3rd Qu.:2016                    3rd Qu.:77.09
Max.   :2021                    Max.   :87.37

               Location     GDP_per_capita                  Income_Group
Afghanistan        :   63   Min.   :   109.6   Low income         :2571
Albania            :   63   1st Qu.:  1298.1   Lower middle income:3432
Algeria            :   63   Median :  4169.8   Upper middle income:2601
Angola             :   63   Mean   : 12033.8   High income        :2886
Antigua and Barbuda:   63   3rd Qu.: 14136.4   NA's               : 720
(Other)            :11340   Max.   :134965.8
NA's               :  555   NA's   :720
```

▶ Show code

```
                           Indicator                   ParentLocation
Life expectancy at age 60 (years):12210   Africa               :3102
Life expectancy at birth (years) :    0   Americas             :2244
                                          Eastern Mediterranean:1452
                                          Europe               :3300
                                          South-East Asia      : 726
                                          Western Pacific      :1386

    Period            Gender     Country_Code
Min.   :2000   Both sexes:4070   Length:12210
1st Qu.:2005   Female    :4070   Class :character
Median :2010   Male      :4070   Mode  :character
Mean   :2010
3rd Qu.:2016
Max.   :2021

              Location.y       Location        Avg_Life_Expectancy
Afghanistan        :   63   Min.   :   109.6   Min.   :69.08
Albania            :   63   1st Qu.:  1298.1   1st Qu.:76.41
Algeria            :   63   Median :  4169.8   Median :78.77
Angola             :   63   Mean   : 12033.8   Mean   :79.01
Antigua and Barbuda:   63   3rd Qu.: 14136.4   3rd Qu.:81.61
(Other)            :11340   Max.   :134965.8   Max.   :89.07
NA's               :  555   NA's   :720
```

▶ Show code

```
   Mode   FALSE    TRUE
logical   11490     720
```

▶ Show code

```
   Mode   FALSE    TRUE
logical   11490     720
```

▶ Show code

```
                           Indicator                   ParentLocation
Life expectancy at age 60 (years):    0   Africa               :2892
Life expectancy at birth (years) :11490   Americas             :2118
```

```
                                    Eastern Mediterranean:1377
                                    Europe                :3150
                                    South-East Asia       : 630
                                    Western Pacific       :1323

    Period            Gender     Avg_Life_Expectancy Country_Code
 Min.   :2001   Both sexes:3830   Min.   :36.60       Length:11490
 1st Qu.:2006   Female    :3830   1st Qu.:64.45       Class :character
 Median :2011   Male      :3830   Median :72.07       Mode  :character
 Mean   :2011                     Mean   :70.39
 3rd Qu.:2016                     3rd Qu.:77.23
 Max.   :2021                     Max.   :87.37

               Location     GDP_per_capita                  Income_Group
 Afghanistan        :   63   Min.   :   109.6   Low income         :2571
 Albania            :   63   1st Qu.:  1298.1   Lower middle income:3432
 Algeria            :   63   Median :  4169.8   Upper middle income:2601
 Angola             :   63   Mean   : 12033.8   High income        :2886
 Antigua and Barbuda:   63   3rd Qu.: 14136.4
 Argentina          :   63   Max.   :134965.8
 (Other)            :11112
```

▶ Show code

```
                           Indicator                   ParentLocation
 Life expectancy at age 60 (years):    0   Africa                :2892
 Life expectancy at birth (years) :11490   Americas              :2118
                                           Eastern Mediterranean:1377
                                           Europe                :3150
                                           South-East Asia       : 630
                                           Western Pacific       :1323

    Period            Gender     Avg_Life_Expectancy Country_Code
 Min.   :2001   Both sexes:3830   Min.   :36.60       Length:11490
 1st Qu.:2006   Female    :3830   1st Qu.:64.45       Class :character
 Median :2011   Male      :3830   Median :72.07       Mode  :character
 Mean   :2011                     Mean   :70.39
 3rd Qu.:2016                     3rd Qu.:77.23
 Max.   :2021                     Max.   :87.37

               Location     GDP_per_capita                  Income_Group
 Afghanistan        :   63   Min.   :   109.6   Low income         :2571
 Albania            :   63   1st Qu.:  1298.1   Lower middle income:3432
 Algeria            :   63   Median :  4169.8   Upper middle income:2601
 Angola             :   63   Mean   : 12033.8   High income        :2886
 Antigua and Barbuda:   63   3rd Qu.: 14136.4
 Argentina          :   63   Max.   :134965.8
 (Other)            :11112
```

4.3 3.3 Storing cleaned data

Describe where you saved the processed file and why.

We saved the processed file under /data/processed because it helps organize the workflow.This structure separates raw data from cleaned and transformed datasets, ensuring the original files remain untouched for reproducibility. This also allows for using the processed data for subsequent analysis, visualization, and reporting steps without needing to repeat the cleaning process.

(Insert code cells – write processed file)

▶ Show code

5 4 Exploratory Data Analysis (EDA)

5.1 4.1 Variable identification

List variable types and their roles.

The merged dataset contains a mix of numeric, factor, and character variables, each serving a distinct analytical purpose. Numeric variables (Period, Avg_Life_Expectancy, GDP_per_capita) represent continuous quantitative data, useful for statistical analysis, calculating averages, and identifying trends over time. Factor variables (Indicator, ParentLocation, Gender, Location.y, Income_Group) categorize data into meaningful groups, allowing comparisons between continents, genders, and economic classifications. Character variables (Country_Code) act as identifiers that link observations across datasets. Together, these variable types enable analysis of relationships between life expectancy, geographical regions, and economic status, supporting both descriptive and inferential insights in later stages of the project.

(Insert code cells – variable listing)

▶ Show code

```
tibble [11,490 × 9] (S3: tbl_df/tbl/data.frame)
 $ Indicator          : Factor w/ 2 levels "Life expectancy at age 60 (years)",..: 2 2 2 2 2 2
2 2 2 2 ...
 $ ParentLocation     : Factor w/ 6 levels "Africa","Americas",..: 1 1 1 1 3 1 3 1 1 1 ...
 $ Period             : num [1:11490] 2021 2021 2021 2021 2021 ...
 $ Gender             : Factor w/ 3 levels "Both sexes","Female",..: 3 3 1 3 3 1 1 2 1 3 ...
 $ Avg_Life_Expectancy: num [1:11490] 48.7 49.6 51.5 51.6 51.8 ...
 $ Country_Code       : chr [1:11490] "LSO" "CAF" "LSO" "SWZ" ...
 $ Location           : Factor w/ 266 levels "Afghanistan",..: 139 42 139 73 219 42 219 139 73
169 ...
 $ GDP_per_capita     : num [1:11490] 1067 492 1067 3966 549 ...
 $ Income_Group       : Factor w/ 4 levels "Low income","Lower middle income",..: 1 1 1 2 1 1
1 1 2 1 ...
```

▶ Show code

```
tibble [11,490 × 9] (S3: tbl_df/tbl/data.frame)
 $ Indicator          : Factor w/ 2 levels "Life expectancy at age 60 (years)",..: 2 2 2 2 2 2
2 2 2 2 ...
 $ ParentLocation     : Factor w/ 6 levels "Africa","Americas",..: 1 1 1 1 3 1 3 1 1 1 ...
 $ Period             : num [1:11490] 2021 2021 2021 2021 2021 ...
 $ Gender             : Factor w/ 3 levels "Both sexes","Female",..: 3 3 1 3 3 1 1 2 1 3 ...
 $ Avg_Life_Expectancy: num [1:11490] 48.7 49.6 51.5 51.6 51.8 ...
```

```
 $ Country_Code       : chr [1:11490] "LSO" "CAF" "LSO" "SWZ" ...
 $ Location           : Factor w/ 266 levels "Afghanistan",..: 139 42 139 73 219 42 219 139 73
169 ...
 $ GDP_per_capita     : num [1:11490] 1067 492 1067 3966 549 ...
 $ Income_Group       : Factor w/ 4 levels "Low income","Lower middle income",..: 1 1 1 2 1 1
1 1 2 1 ...
```

5.2 4.2 Descriptive statistics

Discuss central tendencies, dispersion, outliers.

The analysis of central tendencies and dispersion revealed key patterns in global life expectancy. The mean life expectancy at birth was lower, with a higher standard deviation, compared to life expectancy at age 60 this suggests greater variability and inequality in survival outcomes early in life. In contrast, life expectancy at age 60 showed a higher mean and lower dispersion, indicating that once individuals reach older age, life expectancy becomes more uniform across regions.

Using the custom outlier detection function, outliers were identified only on the lower bound for life expectancy at birth, while no significant outliers appeared at age 60. These lower outliers were concentrated among low and lower-middle income countries, implying that early-life mortality disproportionately affects populations in economically disadvantaged regions. The absence of upper outliers may suggest a natural ceiling to human life expectancy, regardless of socioeconomic factors. This analysis highlights how economic and developmental inequalities strongly shape early-life survival outcomes, which we will further explore in relation to GDP and regional factors in later stages of the project.

(Insert code cells – summarise numeric variables)

▶ Show code

```
[1] 70.39082
```

▶ Show code

```
[1] 8.824191
```

▶ Show code

```
[1] 70.39082
```

▶ Show code

```
[1] 8.824191
```

▶ Show code

Indicator	ParentLocation	Period	Gender
<fct>	<fct>	<dbl>	<fct>
Life expectancy at birth (years)	Eastern Mediterranean	2011	Male
Life expectancy at birth (years)	Eastern Mediterranean	2011	Both sexes
Life expectancy at birth (years)	Americas	2010	Female
Life expectancy at birth (years)	Americas	2010	Both sexes

Indicator	ParentLocation	Period	Gender
<fct>	<fct>	<dbl>	<fct>
Life expectancy at birth (years)	Americas	2010	Male
Life expectancy at birth (years)	Africa	2010	Male
Life expectancy at birth (years)	Africa	2009	Male
Life expectancy at birth (years)	Africa	2009	Both sexes
Life expectancy at birth (years)	Africa	2009	Male
Life expectancy at birth (years)	Africa	2008	Male

1-10 of 66 rows | 1-4 of 9 columns Previous **1** 2 3 4 5 6 7 Next

▶ Show code

Indicator	ParentLocation	Period	Gender
<fct>	<fct>	<dbl>	<fct>
Life expectancy at birth (years)	Eastern Mediterranean	2011	Male
Life expectancy at birth (years)	Eastern Mediterranean	2011	Both sexes
Life expectancy at birth (years)	Americas	2010	Female
Life expectancy at birth (years)	Americas	2010	Both sexes
Life expectancy at birth (years)	Americas	2010	Male
Life expectancy at birth (years)	Africa	2010	Male
Life expectancy at birth (years)	Africa	2009	Male
Life expectancy at birth (years)	Africa	2009	Both sexes
Life expectancy at birth (years)	Africa	2009	Male
Life expectancy at birth (years)	Africa	2008	Male

1-10 of 66 rows | 1-4 of 9 columns Previous **1** 2 3 4 5 6 7 Next

▶ Show code

0 rows | 1-6 of 9 columns

▶ Show code

Indicator	ParentLocation	Period	Gender
<fct>	<fct>	<dbl>	<fct>
Life expectancy at birth (years)	Eastern Mediterranean	2011	Male
Life expectancy at birth (years)	Eastern Mediterranean	2011	Both sexes
Life expectancy at birth (years)	Americas	2010	Female
Life expectancy at birth (years)	Americas	2010	Both sexes
Life expectancy at birth (years)	Americas	2010	Male
Life expectancy at birth (years)	Africa	2010	Male
Life expectancy at birth (years)	Africa	2009	Male
Life expectancy at birth (years)	Africa	2009	Both sexes
Life expectancy at birth (years)	Africa	2009	Male
Life expectancy at birth (years)	Africa	2008	Male

1-10 of 66 rows | 1-4 of 9 columns Previous **1** 2 3 4 5 6 7 Next

▶ Show code

Indicator	ParentLocation	Period	Gender
<fct>	<fct>	<dbl>	<fct>
Life expectancy at birth (years)	Eastern Mediterranean	2011	Male
Life expectancy at birth (years)	Eastern Mediterranean	2011	Both sexes
Life expectancy at birth (years)	Americas	2010	Female
Life expectancy at birth (years)	Americas	2010	Both sexes
Life expectancy at birth (years)	Americas	2010	Male
Life expectancy at birth (years)	Africa	2010	Male
Life expectancy at birth (years)	Africa	2009	Male
Life expectancy at birth (years)	Africa	2009	Both sexes
Life expectancy at birth (years)	Africa	2009	Male
Life expectancy at birth (years)	Africa	2008	Male

1-10 of 66 rows | 1-4 of 9 columns Previous **1** 2 3 4 5 6 7 Next

▶ Show code

0 rows | 1-6 of 9 columns

▶ Show code

ParentLocation	Income_Group	n
<fct>	<fct>	<int>
Africa	Low income	35
Africa	Lower middle income	26
Americas	Lower middle income	3
Eastern Mediterranean	Low income	2

4 rows

▶ Show code

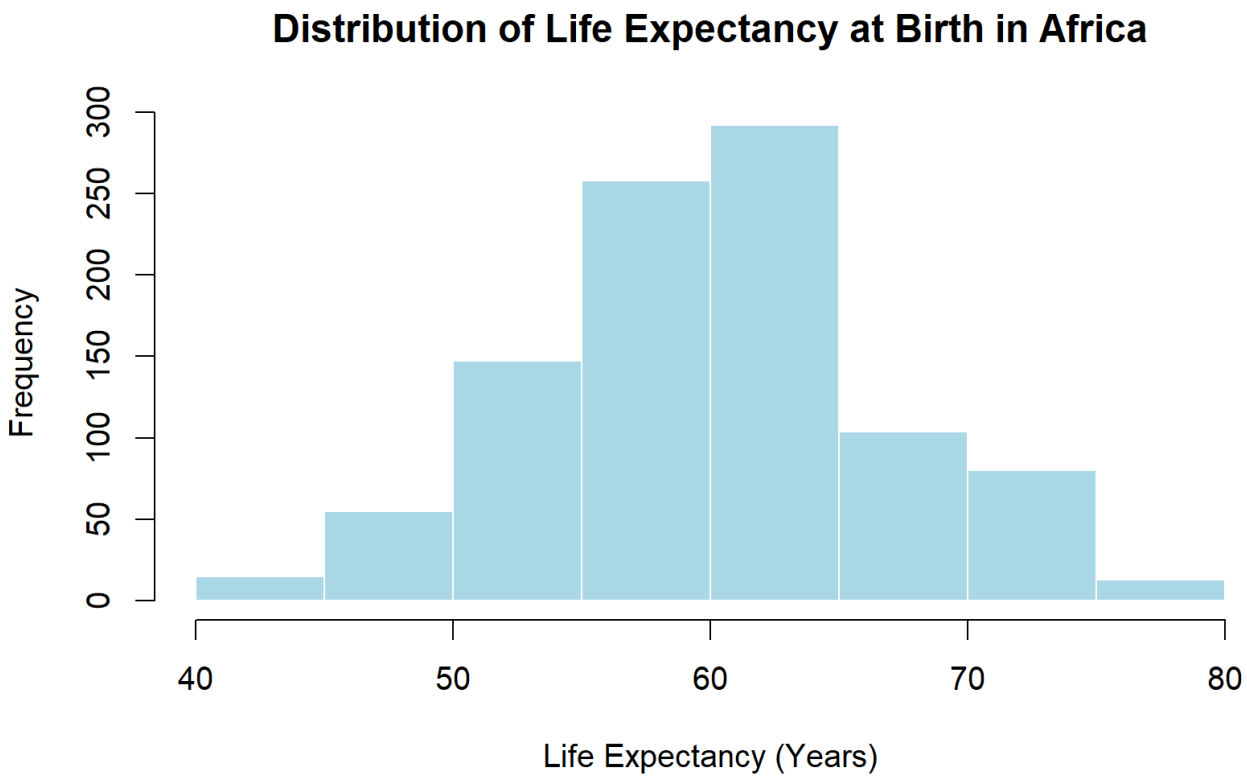
5.3 4.3 Numerical analysis

Discuss histograms and boxplots

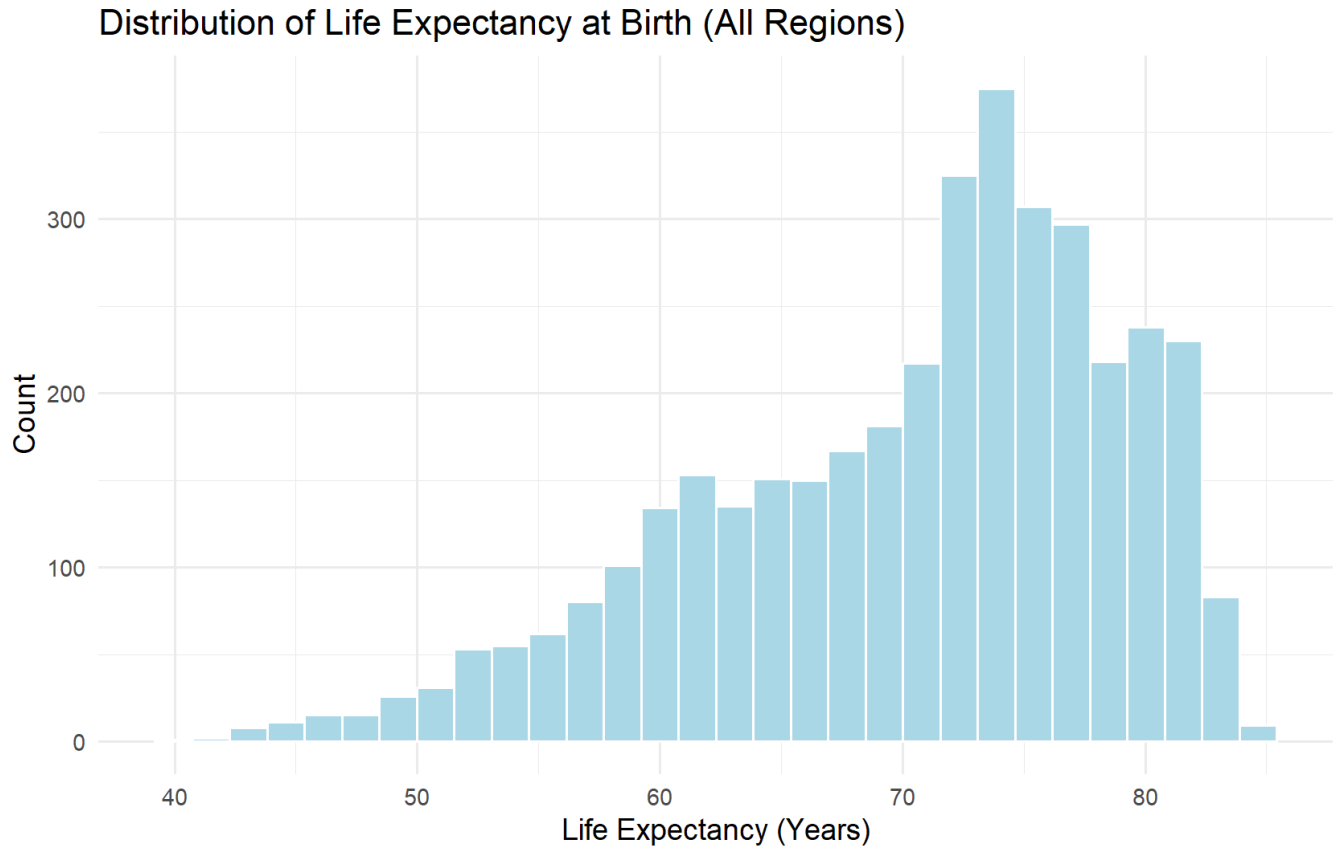
We will be using histogram to understand the overall pattern and shape of life expectancy values, observe where most countries fall and how the extremes behave. We also make use of boxplots to see which regions or groups perform better or worse in terms of life expectancy, and where the most extreme cases occur.

(Insert code cells - histograms and baxplots)

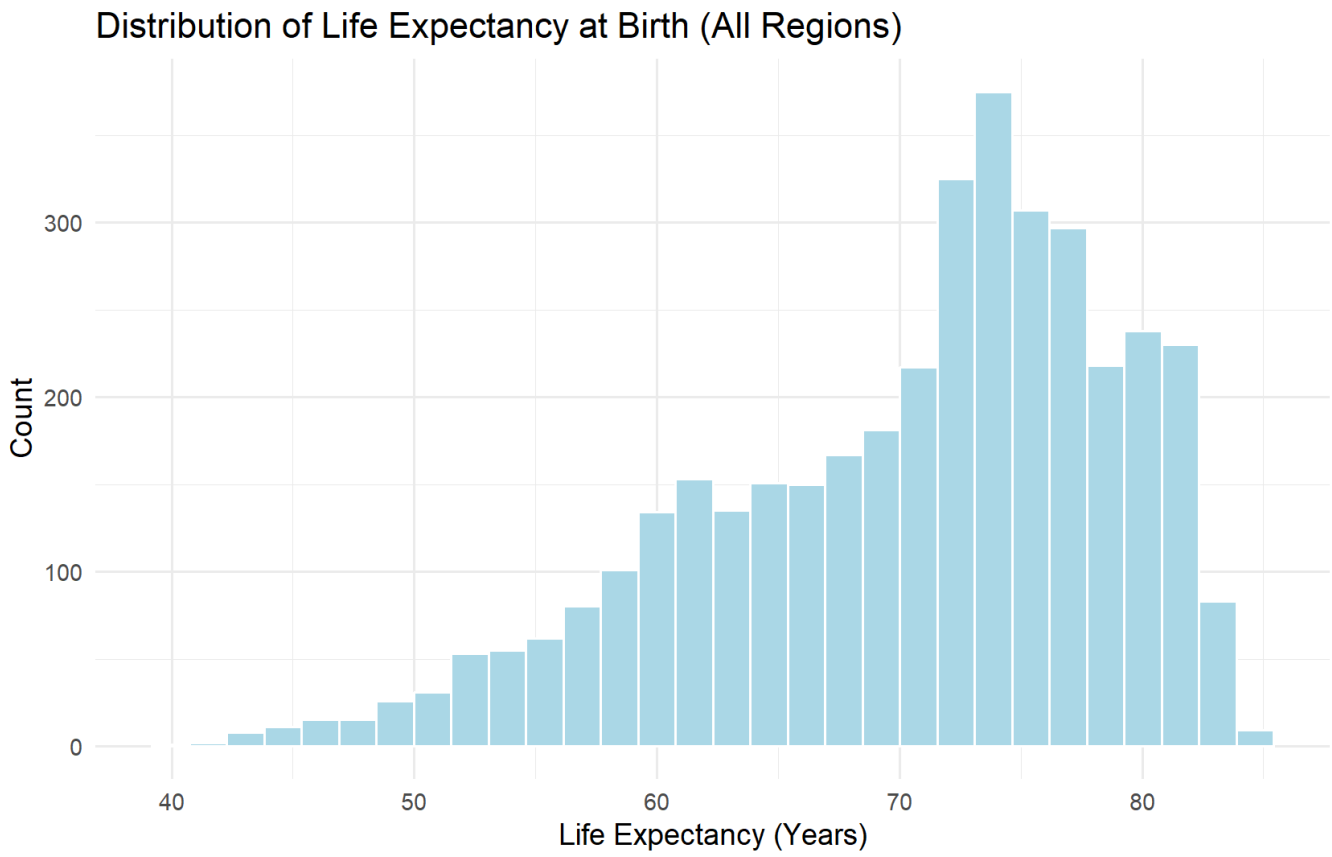
▶ Show code



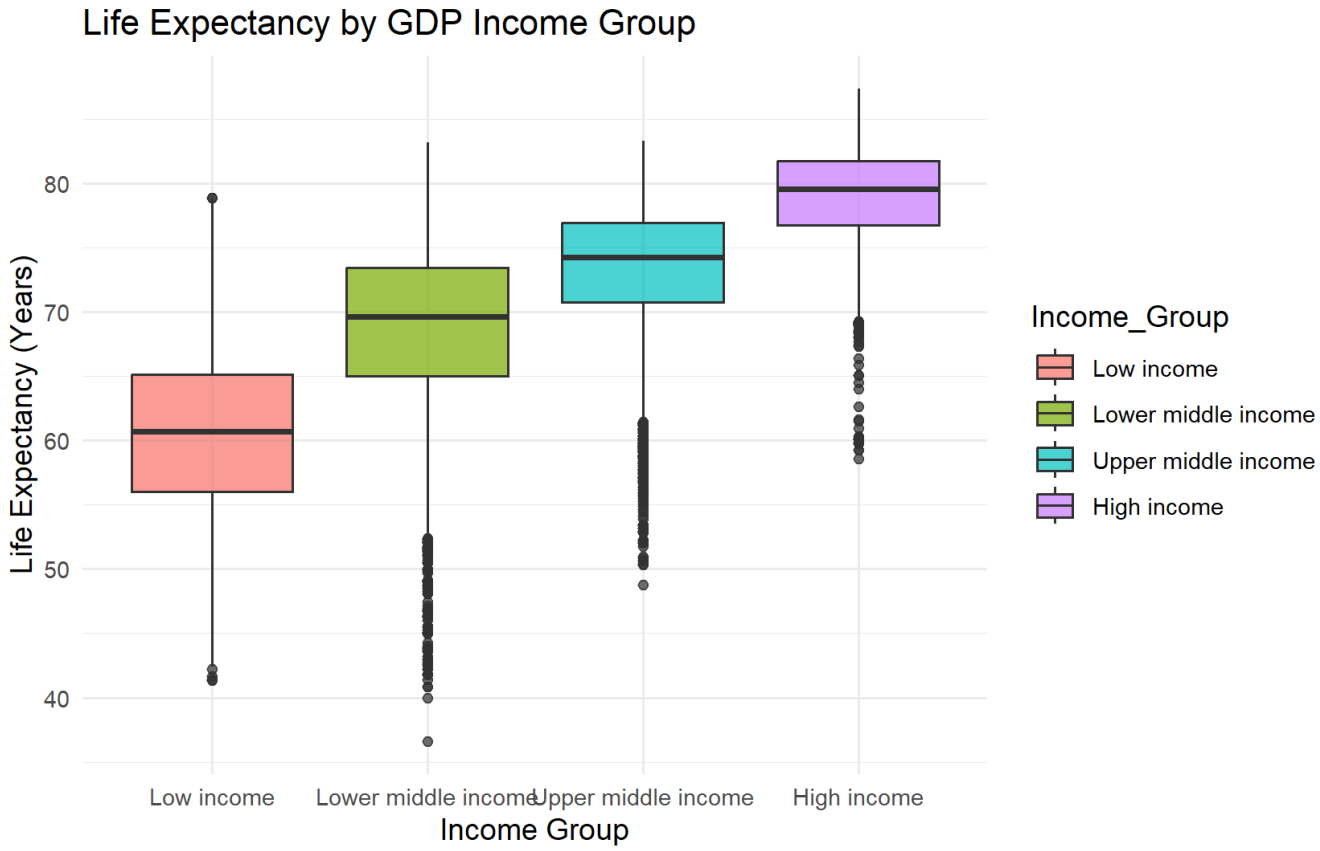
▶ Show code



▶ Show code



▶ Show code



5.4 4.4 Categorical analysis

Discuss frequency distributions or group counts.

The frequency distributions and group counts provided valuable insight into how the data is structured across key dimensions. When grouped by income level, the majority of records were found within the lower-middle income category, followed by low-income countries which highlights a significant portion of the world still faces economic constraints that may influence health outcomes. Regional counts showed the largest representation from Africa, Europe, and the Americas, which aligns with the number of countries in these regions and suggests a fairly balanced dataset geographically.

Gender counts confirmed that data were available across all gender categories (“Both sexes,” “Male,” and “Female”), ensuring completeness for later comparative analysis. When examining the interaction between income group and WHO region, a clear socioeconomic pattern emerged: most African countries fall into low or lower-middle income groups, while European countries are predominantly high-income.

This uneven economic distribution is a critical contextual factor when interpreting life expectancy differences. Finally, the trend plot of life expectancy over time revealed that Africa consistently trails behind other regions, which could significantly influence the global average. These frequency analyses help establish a foundational understanding of the dataset’s composition and set the stage for deeper regional and economic comparisons in later stages of the project.

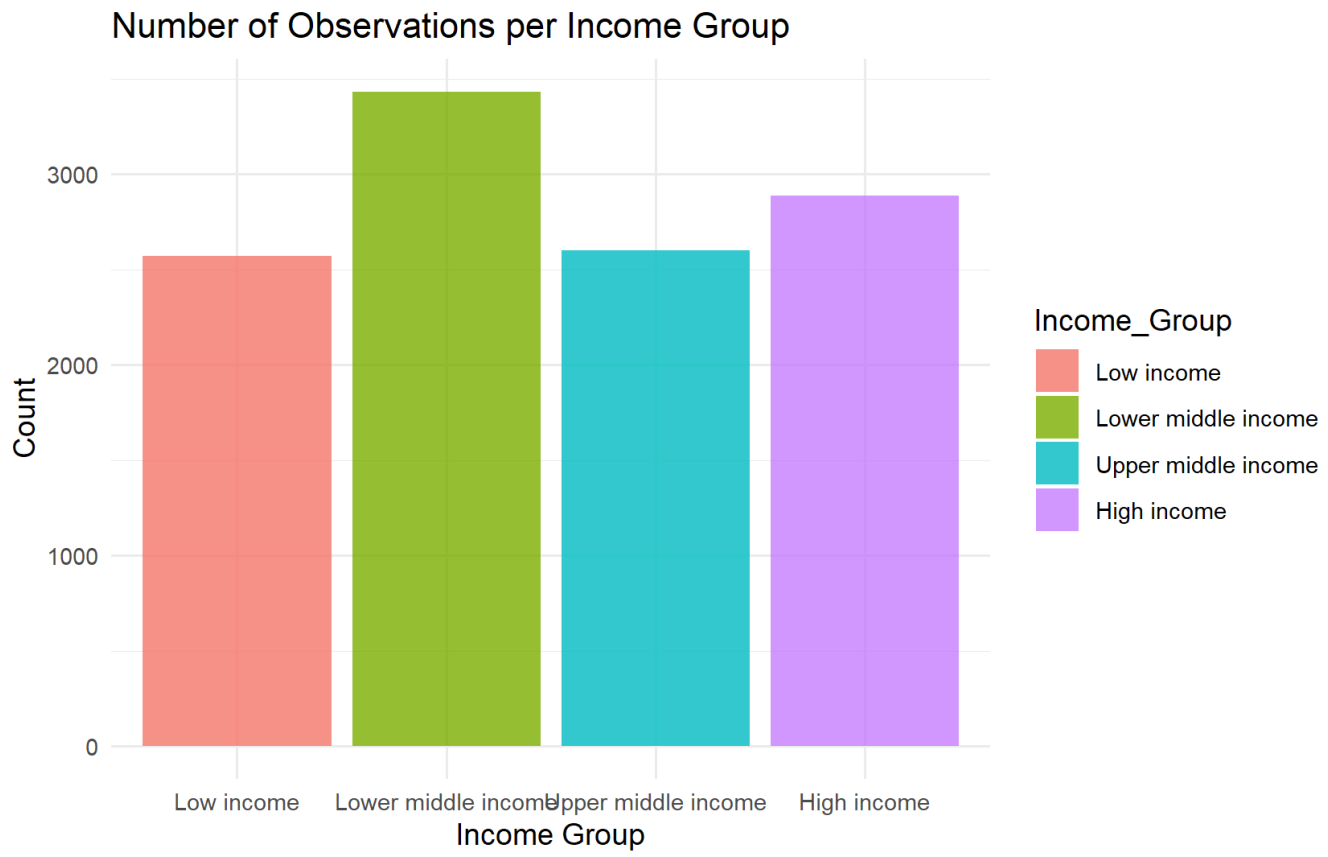
(Insert code cells – counts or plots)

▶ Show code

Income_Group	**n**
<fct>	<int>
Lower middle income	3432
High income	2886
Upper middle income	2601
Low income	2571

4 rows

▶ Show code

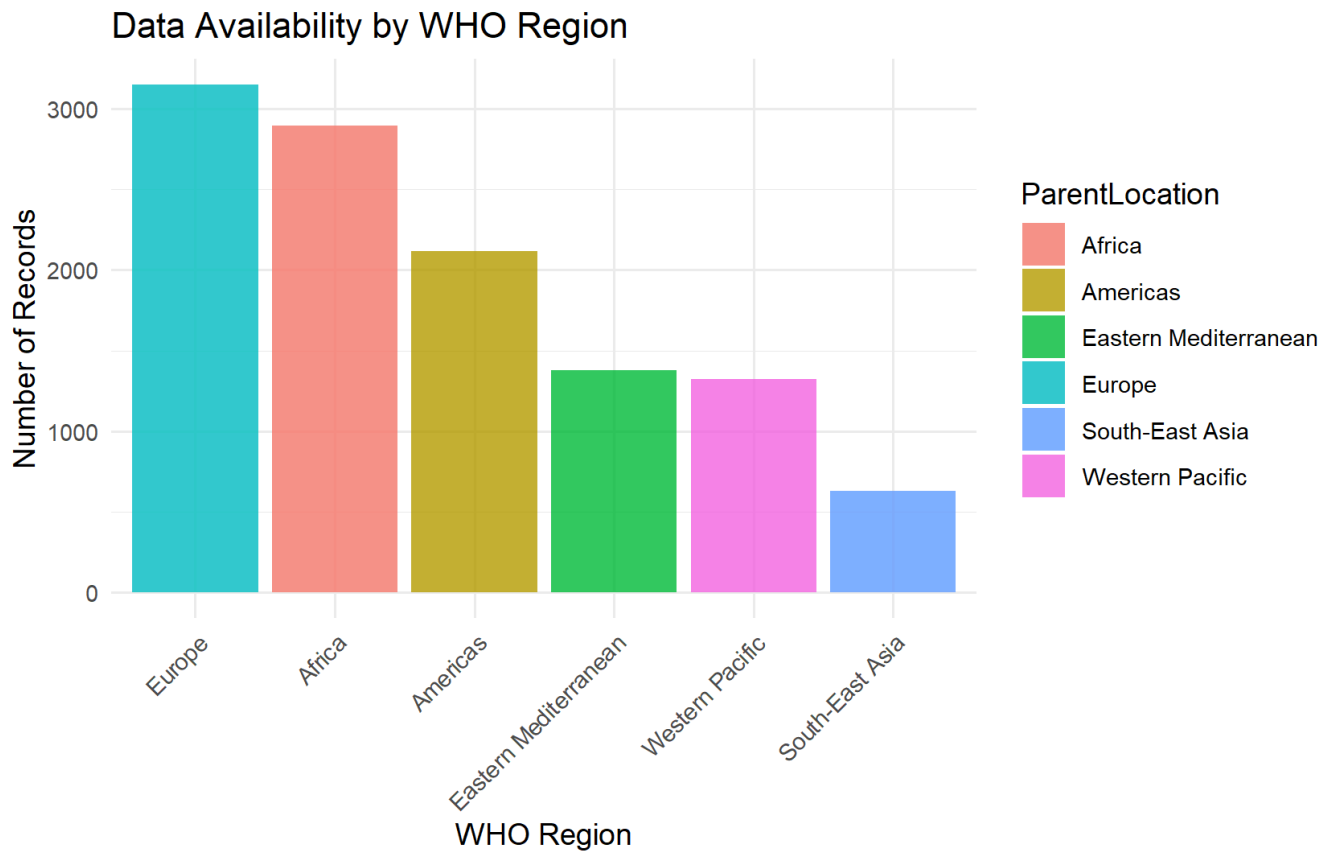


▶ Show code

ParentLocation	n
<fct>	<int>
Europe	3150
Africa	2892
Americas	2118
Eastern Mediterranean	1377
Western Pacific	1323
South-East Asia	630

6 rows

▶ Show code

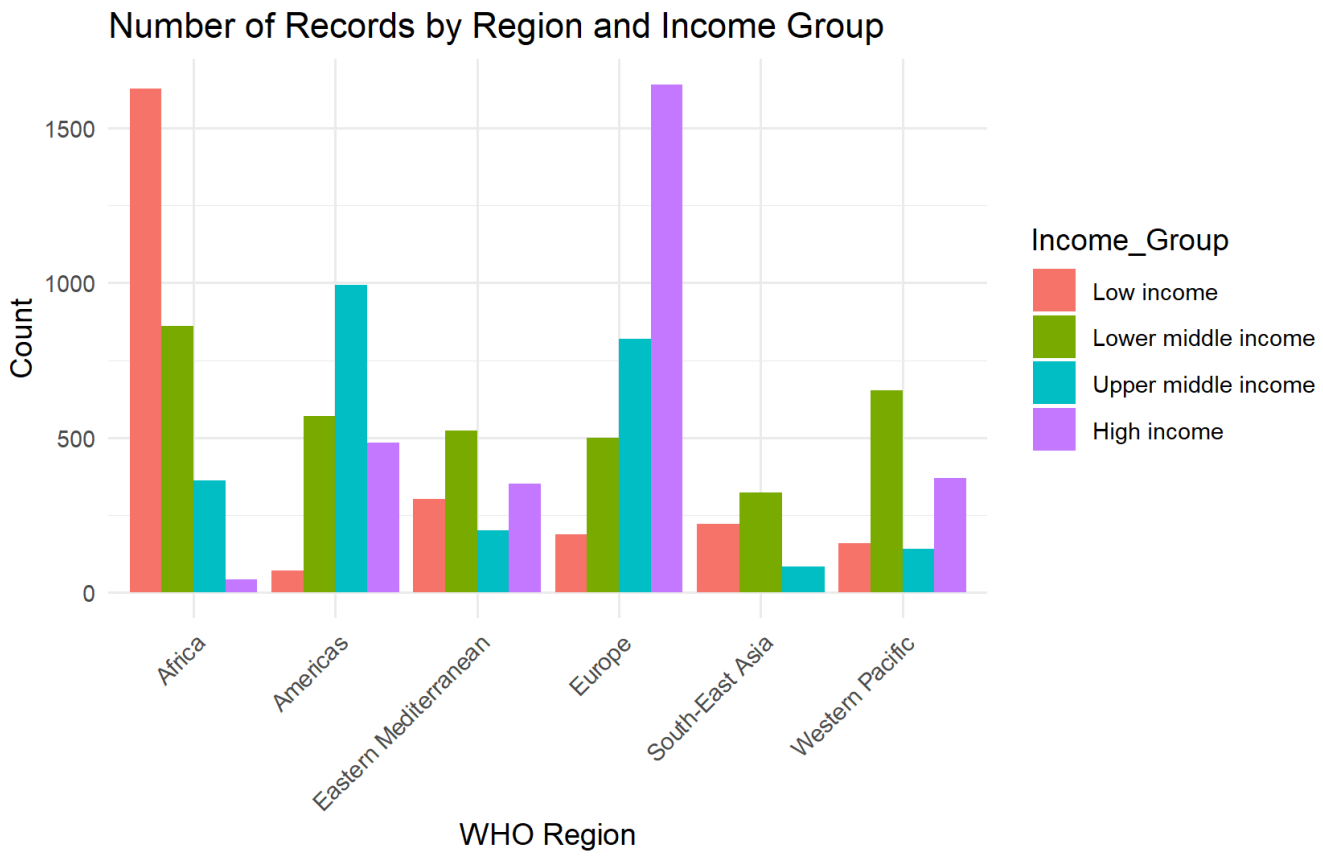


▶ Show code

Gender	n
<fct>	<int>
Both sexes	3830
Female	3830
Male	3830

3 rows

▶ Show code

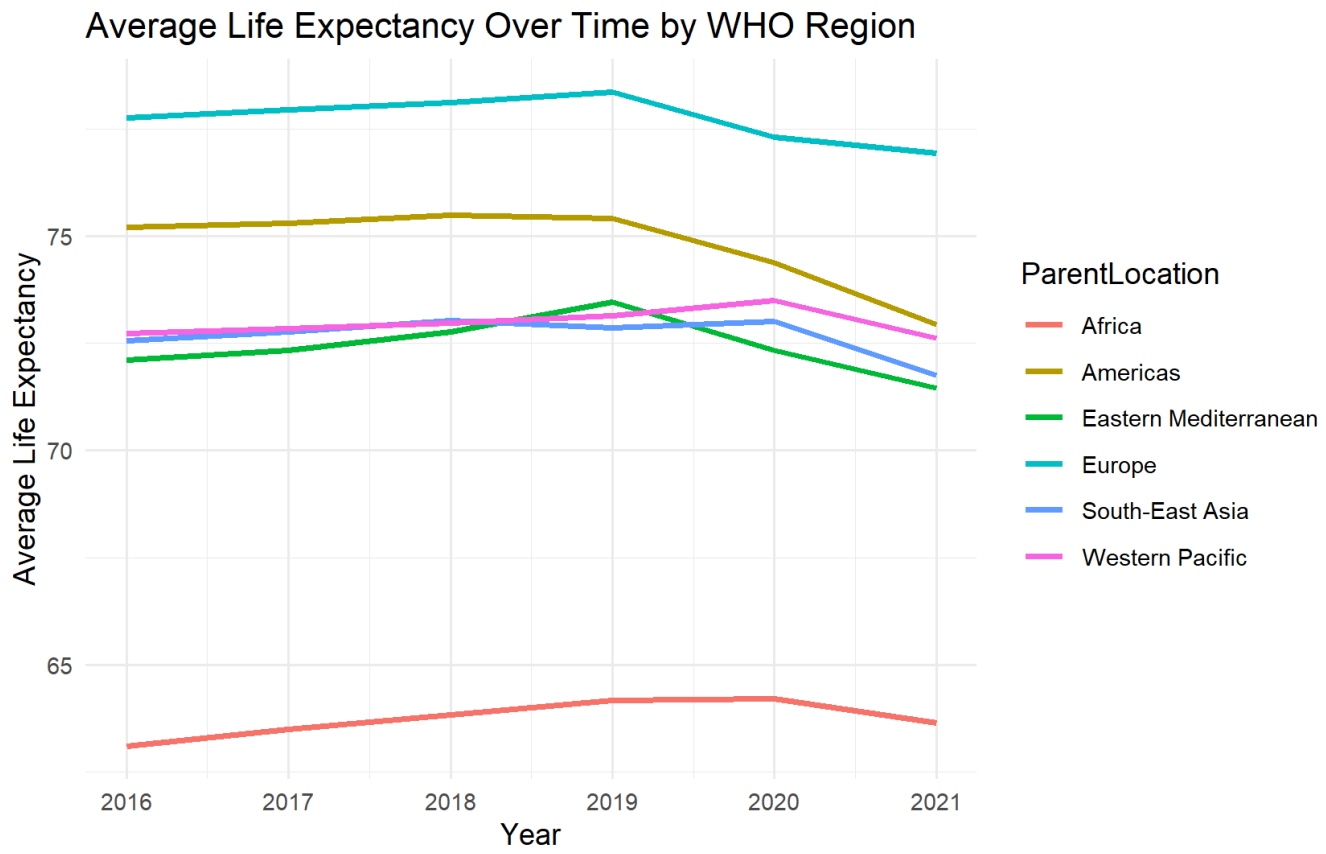


▶ Show code

Indicator	ParentLocation	Period	Gender
<fct>	<fct>	<dbl>	<fct>
Life expectancy at birth (years)	Africa	2021	Male
Life expectancy at birth (years)	Africa	2021	Male
Life expectancy at birth (years)	Africa	2021	Both sexes
Life expectancy at birth (years)	Africa	2021	Male
Life expectancy at birth (years)	Eastern Mediterranean	2021	Male
Life expectancy at birth (years)	Africa	2021	Both sexes
Life expectancy at birth (years)	Eastern Mediterranean	2021	Both sexes
Life expectancy at birth (years)	Africa	2021	Female
Life expectancy at birth (years)	Africa	2021	Both sexes
Life expectancy at birth (years)	Africa	2021	Male

1-10 of 3,246 rows | 1-4 of 9 columns Previous **1** 2 3 4 5 6 … 325 Next

▶ Show code



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5.5 4.5 Interpretation

Write short commentary linking patterns to your project context.

Wealth and Health: The boxplot and scatter plot reveal a simple, powerful pattern: the higher a country's GDP, the longer its citizens tend to live. This confirms our main hypothesis and provides a solid foundation for exploring global health inequality.

Regional Disparities: The line graph highlights significant gaps between different parts of the world. For example, average life expectancy in Africa is consistently lower than in Europe. This visualization allows us to see which regions are lagging and how these inequalities change over time.

Crisis Points: The outliers in the data are not just statistics; they represent countries facing severe crises. These extreme cases are crucial for understanding the real-world impact of the health and economic disparities we are investigating.

6 5 Documentation and Reflection

Capture reasoning, challenges, and decisions.

The boxplots and histograms clearly demonstrate a consistent trend that countries with higher GDP per capita exhibit higher average life expectancy. This supports the central hypothesis that economic prosperity directly translates into better health outcomes through improved healthcare access, nutrition, and living conditions. The pattern is most evident when countries are grouped by income level: the higher the GDP group, the higher the median life expectancy and the smaller the spread.

1.Regional Disparities: The line graphs highlight persistent regional gaps in life expectancy over time. African countries consistently display the lowest averages, while Europe and the Western Pacific regions maintain the highest. This confirms that global health inequality remains strongly geographic — closely tied to regional wealth and development patterns.

These visualisations provide the foundation for deeper investigation into why some regions continue to lag behind**, despite overall global progress.

2.Crisis Points: The outlier analysis revealed that outliers exist only on the lower bound, representing countries where life expectancy is exceptionally low. These correspond primarily to low- and lower-middle-income African countries, suggesting severe inequality in early-life mortality. The absence of upper-bound outliers may imply a natural ceiling to human longevity across all regions — reinforcing the idea that inequality is driven by deprivation rather than exceptional success.

#Key Decisions:

1.Dataset Fusion: We merged WHO Life Expectancy and World Bank GDP datasets to link health outcomes directly with economic conditions across 194 countries. This enabled integrated analysis of global health inequality through both socioeconomic and regional dimensions.

2.Metric Separation: We split the analysis into two complementary datasets life expectancy at birth and life expectancy at age 60. This allowed us to capture two layers of inequality; Early-life mortality and developmental health conditions (at birth), and Later-life longevity and aging outcomes (at 60). The contrast between these metrics revealed that disparities are primarily concentrated at birth.

3.Data Reshaping and Categorisation: Transforming the GDP data into long format with `pivot_longer()` and creating the `Income_Group` variable were essential for meaningful comparison and visualisation. This step enabled the use of grouped boxplots and time-series plots, making regional and economic inequalities more interpretable.

Key Decisions Dataset Fusion: We merged WHO Life Expectancy and World Bank GDP data to directly link health outcomes with economic status.

Metric Separation: We split the analysis into “at birth” and “at age 60” to capture different aspects of health inequality—from infant mortality to adult longevity.

Data Reshaping: Transforming the GDP data to a long format and creating Income_Group categories were essential for carrying out time-series analysis and creating clear, comparative plots like the boxplot.

#Challenges:

1.Data Integration The biggest challenge was harmonising country codes and variable names between the WHO and World Bank datasets. Without standardised schemas, merging required careful alignment of join keys (`Country_Code` and `Period`) and validation to ensure accuracy.

2.Missing Values Although missing GDP values made up only about 5.8 % of records, they still affected completeness. For this analysis we opted to remove rows containing NA values to maintain data consistency, but future modelling will require more robust strategies (e.g., interpolation or imputation).

3.Outlier Identification Developing a custom `find_outliers()` function to return both upper and lower bounds was initially complex. This function revealed an insightful pattern — all significant outliers occur only

on the lower bound, underscoring extreme inequality in health outcomes for a small group of nations.

4.Interpretation Linking statistical outliers to meaningful real-world contexts (such as conflict, poverty, or public health crises) required external reasoning beyond the dataset itself.

#Future Scope:

1.Add More Variables Include additional socioeconomic and health-related indicators (e.g., healthcare expenditure, education level, or infant mortality rate) to better explain the underlying causes of inequality.

2.Investigate Outliers Further Conduct targeted case studies on countries identified as lower-bound outliers to understand contextual drivers such as conflict, epidemics, or governance issues.

3.Apply Statistical Modelling Extend the analysis from exploratory data analysis (EDA) to inferential modelling — for instance, by running regression analyses to quantify the effect of GDP and region on life expectancy.

4.Visual and Interactive Extensions Build a dashboard (e.g., with `shiny` or `plotly`) for interactive exploration of regional trends and income group disparities over time.

6.1 5.1 What Assumptions Did You Make?

1.Data Source Reliability: The WHO and World Bank data were assumed to be accurate, reliable, and collected using standardised global methodologies.

2.GDP as a Proxy for Development: We assumed GDP per capita is a valid proxy for living standards and access to essential services, even though it does not capture inequality within countries.

3.Consistency of Definitions: Life expectancy and GDP per capita were assumed to be measured consistently across all countries and years, allowing direct comparison.

4.Meaningful Groupings: By categorising countries into income groups, we assumed that countries within the same bracket share comparable socioeconomic conditions relevant to health analysis.

6.2 5.2 What was easy or hard?

Relatively Easy :

1.Data Loading and Inspection: Using `tidyverse` functions such as `read_csv()`, `glimpse()`, and `summary()` made initial exploration straightforward.

2.Reshaping and Cleaning: `pivot_longer()` and `mutate()` simplified the conversion of GDP data into a usable long format.

3.Visualisation: Creating histograms, boxplots, and time-series plots using `ggplot2` was intuitive and effective for validating early insights.

Relatively Hard:

1.Data Integration: Aligning datasets from two distinct institutions required careful type conversion and key matching.

2.Outlier Logic: Writing and validating the custom function to correctly isolate both upper and lower outliers took multiple iterations.

3.Interpretation: Translating statistical patterns into meaningful real-world implications (e.g., inequality, health crises) required critical reasoning.

4.Handling Missing Data: Dealing with partial GDP coverage demanded trade-offs between data completeness and accuracy.

5.Developing a Strategy for Missing Data: A significant challenge, which will become more critical in the next phase, is handling the missing GDP values for certain countries and years. Whilst our initial plots can accommodate these gaps, deciding on a robust and statistically sound strategy for dealing with them (e.g., removal, imputation) before formal modelling is a non-trivial problem that requires careful consideration to avoid introducing bias.

6.3 5.3 What would you do differently next time?

1.Pre-filter Regional Aggregates: Remove non-country regional summary rows (e.g., “Europe” or “Africa”) before merging to reduce NA values and simplify cleaning.

2.Standardise Join Keys Early: Ensure that `Country_Code` and `Period` columns are changed to the same data type immediately after data import to streamline merging.

3.Implement Missing Value Strategy: Use statistical imputation or interpolation rather than simple row deletion to retain more data for modelling.

7 6 Good Practice Reminders

Checklist and reproducibility notes.

- Comments in code
- Relative paths
- Push and pull discipline
- Consistent naming conventions

8 7 Group Notes and Variations

Describe context-specific adaptations or creative extensions. n the project, the group introduced several context-specific adaptations and creative extensions to tailor the analysis to global health inequality.

Contextual Adaptation of Datasets: Instead of relying on a single source, we merged two authoritative datasets; WHO Life Expectancy and World Bank GDP per capita, to capture both health and economic dimensions of development. This fusion allowed us to explore the relationship between wealth and health outcomes across 194 countries. We specifically focused on two indicators of life expectancy; at birth and at age 60, to differentiate between early-life mortality and later-life longevity, providing more understanding of inequality.

Reshaping for Comparative Analysis: We creatively transformed the GDP dataset using pivot_longer() to convert wide format year columns into a “Year” variable enabling longitudinal analysis. This restructure

supported the direct comparison with the WHO life expectancy dataset by aligning them both on Country_Code and Period. Categorizing countries into World Bank income groups (Low, Lower-Middle, Upper-Middle, High) added a meaningful socioeconomic lens for comparative visualizations.

Custom Analytical Enhancements: We developed a custom function, find_outliers(), to detect and inspect both lower and upper outliers. This helped uncover that outliers were concentrated only on the lower bound of life expectancy and thus revealing disparities among low-income countries. The function was designed for reusability making it a versatile analytical tool for future projects.

Regionally Focused Visual Exploration: To better contextualize global inequalities, we generated region-specific histograms and line plots (e.g. for Africa) that highlighted geographic disparities. These visual extensions allowed us to interpret outliers not just as statistical anomalies but also as reflections of real-world socioeconomic and public health challenges.

Practical Workflow Enhancements: By saving processed datasets under data/processed, we established a clear and reproducible workflow separating raw, intermediate, and final data. This structure promotes collaboration and ensures that subsequent analysis, visualization, and modelling steps can proceed efficiently without redundant preprocessing.

Creative Interpretation: Beyond statistical summaries, our interpretations framed the findings in the context of global inequality, regional development patterns, and early-life mortality. This approach transforms data analysis into a tool for understanding global social issues linking numbers to real human outcomes.

9 8 Final Checkpoint Before Submission

Tick items off to ensure readiness.

- Dataset in /data/raw/
- Cleaned dataset in /data/processed/
- HTML output in /outputs/
- Reflection written
- Changes committed and pushed

Week 3 – Understanding Your Data: Provenance, Quality & Risks

AUTHOR
Group Name / Members

PUBLISHED
November 17, 2025

Overview

> **What this file is for**
>
> This week's focus is more on **reflection and documentation**, and less or not on coding.
> You will describe where your data comes from, how trustworthy it is, and what risks or limitations it may carry.

---

1 Dataset Overview

A short paragraph (3–5 sentences) describing:

- What your dataset is about
  Our Life Expectancy dataset provides information on the Average life expectancy at Birth and at 60 of Males, Females and both sexes in 185 countries between 2000 and 2021. Our second dataset contains the GDP of over 200 countries dating from 1964 to 2024.
- Who published it
  World Health Organisation(WHO) for the Life Expectancy data and World Bank Group(WBG) for the Gross Domestic Product(GDP) data of the countries.
- When and how you accessed it
  We downloaded the data for Life Expectancy on the 14th October 2025 and the GDP data on the 15th October 2025. We accessed the life expectancy data from the link provided when choosing research topics. On the other hand the GDP data was obtained by researching various sources of GDP information and deciding the World Bank data was the most complete and reliable informartion.
- Why it fits your project idea Our chosen research topic was to observe the differences in Life expectancy between high and low income countries and find reasons why the gaps exist. The GDP data allowed us to group countries by "High income", "low middle income" , "high middle income" and "high income" countries as outlined by the World Bank standard. The life expectancy data gave us access to data for Average life expectancy for different genders and regions. With a combination of these two datasets, we will be able to analyse and visualize the regional inequality in terms of life expectancy.

---

2 Provenance Summary

Use the table below to record evidence and assessments about your dataset. You can edit or expand it as needed. Please provide short support statement (1-2 lines each) for your verditcs.

Criterion	Evidence Checked	Verdict
Origin & Authority	Who created the dataset? What was their purpose or	✅ / ⚠️ / ❌

Criterion	Evidence Checked	Verdict
	methodology?	

Origin & Authority | Who created the dataset? What was their purpose or methodology? - WHO estimates the global health parameters and WBG focuses on world development indicators| ✅ |Both datasets come from the primary, globally recognised authorities on Health and Economic data, their credibility is excelent.Both use official country reports and are official sources that use standardised global methods for compiling data.|
Coverage & Completeness | Time span, geography, variables, sample size, missingness | ✅ / ⚠️ / ❌ |
Coverage & Completeness | Time span, geography, variables, sample size, missingness - Global coverage and different time spans with GDP from the year 1960 and LE from the year 2000 | ⚠️ | Global and gender coverage is good from both datasets as up to 185 countries are covered and male, female and both sexes are present in analysis. The different starting years limit the merged analysis from the two datasets and the missing GDP values create analytical gaps.|
Timeliness & Stability | Last update date, update frequency, schema consistency | ✅ / ⚠️ / ❌ |
Timeliness & Stability | Last update date, update frequency, schema consistency - Data is the latest available from the WHO and WBG as sources are actively maintained. | ✅ | Both datasets are actively managed and updated frequently by their respective institutions as the WHO's data is recent to 2021 and the WBG's data is recent to 2024, ensuring a high degree of current and structural stability.|
Format & Access | CSV/Excel/JSON/API? Can you reproduce the import path? | ✅ / ⚠️ / ❌ |
Format & Access | CSV/Excel/JSON/API? Can you reproduce the import path? - CSV files have a reproducible load path using read_csv() in R. | ✅ | Both are accessible from public portals as CSV files, This makes the import process simple and fully reproducible for subsequent analysis.|
Documentation & Versioning | Data dictionary, changelog, DOI/version number | ✅ / ⚠️ / ❌ |
Documentation & Versioning | Data dictionary, changelog, DOI/version number - Both WHO's global health estimates and World Bank's development indicators provide extensive online methodology and data dictionaries | ✅ |
There is clear and comprehensive documentation which is publicly available and clarifies the sources, definitions, and estimation methods used for their data collection.

License & Attribution | License type, URL, and required credit statement | ✅ / ⚠️ / ❌ |
License & Attribution | License type, URL, and required credit statement - World Bank and WHO generally use open data licenses for any user needing access to data | ✅ | Licensing is open and permits reuse for any academic or professional work with mandatory citations of the original source.
Sensitivity & Ethics | Contains or could infer personal or sensitive information? | ✅ / ⚠️ / ❌ |
Sensitivity & Ethics | Contains or could infer personal or sensitive information? - Data is aggregated | ✅ | The data is aggregated at the national level and contains no personal, identifiable, or ethically sensitive individual information

2.1 Understanding Each Criterion

- **Origin & Authority:** Identify who created or published the dataset, and why it exists. Check for official sources or credible research institutions.
- **Coverage & Completeness:** Consider whether the data spans a sufficient period, geography, and number of observations for your analysis. Missing years or regions may signal gaps.
- **Timeliness & Stability:** When was it last updated? Has the structure changed over time? Outdated or unstable sources may reduce reliability.

- **Format & Access:** Ensure the format (CSV, JSON, etc.) can be opened reproducibly with R/Python. Record any URLs or APIs used.
- **Documentation & Versioning:** Look for accompanying documentation (README, codebook, changelog) or version numbers that clarify structure.
- **License & Attribution:** Record the license name (e.g., CC BY 4.0, OGL v3.0) and link. Copy the required attribution statement if available.
- **Sensitivity & Ethics:** Note whether the data includes personal, identifiable, or ethically sensitive information. If yes, explain how you'll handle it responsibly.

2.2 Licensing Details

Fill in the fields below using information from your dataset's source page. Copy the exact text if provided.

License Name: World Bank (GDP)- Creative Commons Attribution 4.0 (CC BY 4.0) WHO (Life Expectancy) - Creative Commons Attribution-NonCommercial-ShareAlike 3.0 IGO (CC BY-NC-SA 3.0 IGO)

License Link: World Bank - https://creativecommons.org/licenses/by/4.0/ WHO - https://creativecommons.org/licenses/by-nc-sa/3.0/igo/

Attribution Text (if provided): World Bank (GDP)- You must give appropriate credit , provide a link to the license, and indicate if changes were made . You may do so in any reasonable manner, but not in any way that suggests the licensor endorses you or your use. WHO (Life Expectancy) - You must give appropriate credit , provide a link to the license, and indicate if changes were made . You may do so in any reasonable manner, but not in any way that suggests the licensor endorses you or your use.

Notes on Limitations (e.g., Non-Commercial):
World Bank(GDP) - You do not have to comply with the license for elements of the material in the public domain or where your use is permitted by an applicable exception or limitation . No warranties are given. The license may not give you all of the permissions necessary for your intended use. For example, other rights such as publicity, privacy, or moral rights may limit how you use the material. WHO (Life Expectancy)- When the Licensor is an intergovernmental organization, disputes will be resolved by mediation and arbitration unless otherwise agreed. You do not have to comply with the license for elements of the material in the public domain or where your use is permitted by an applicable exception or limitation . No warranties are given. The license may not give you all of the permissions necessary for your intended use. For example, other rights such as publicity, privacy, or moral rights may limit how you use the material.

> **Tip**
>
> If you cannot find a license, state:
> *"No explicit license visible; flagged for clarification or restricted use."*

---

3 Early Risks Reflection

Write about **150 words** identifying key risks or concerns, such as:

- Missing data or uneven coverage

There was a large number of columns with data that had absolutely no information in the life expectancy data. The GDP data was missing values for some countries in certain years but majority of the missing values were outside of the range we intended to research on.

- Potential bias or lack of representativeness

There is data on 184 countries which is a very representative sample. We have information on all of the major continents. We may find a bias in how we have not considered other forms of Life expectancy such as healthy Average Life Expectancy .

- Ethical or licensing uncertainties
  Our primary ethical risk is the systematic exclusion of the term gender in favour of sex. These terms are incorrectly used interchangeably which caused us to incorrectly label the “sex” field “gender”. Since the WHO data consists of sex data, we can only analyse information on life expectancy of the two biological sexes. This means our final results will under represent the different genders. We cannot provide complete gender information as we do not have access to such data therefore our only solution is to rename the gender column to sex so as to provide some completeness to the data. We do note that we would have preferred to have gender information to improve our level of representation. Data collection is not a neutral act; it is a form of power. The choices made about what to display (and what not to display) have real-world consequences.

Licensing risks emerge as the WHO data’s “Non-Commercial” licensing restricts our final merged dataset. This limits the reproducibility of our work, as others cannot legally reuse our final merged_table_Birth.csv for any commercial purpose, a critical limitation we must clearly state.

- How these issues might affect later analysis

As a group we have agreed to add HALE as an extra variable to see how previous years of unhealthy living may have influenced life expectancy figures. We have also chosen to change the gender “variable” to “sex”

> *Tip:* Think “If someone reused our results, what could mislead them?”

---

4 (Optional) API Comparison

If applicable, you may compare data access via direct download vs API.
Below is an optional R code snippet to illustrate how APIs differ from static downloads.

▶ Show code

Summarise your observations (e.g., freshness, column structure, metadata differences).

---

5 (Optional) Scouting / Alternative Data

List any other datasets you explored (name, link, why rejected or unsuitable). These could be the datasets that are somehow comparable to what you are currently using.
A short bullet list is enough.

Dataset / Source	Relation to Main Dataset	Outcome / Issues

Week 4 - Statistics: Distributions & First Model

Lecturer: Dr Jon Cardoso-Silva
Course Code: MATH225_WP101_2024S1_01

Semester 1
November 27, 2024

Overview

Click this file to view

This course covers Distributions, Tests, Math. Measured model. PR 225 the fundamental variables. Probable distribution shapes Histograms - Normal and bell. The probable linear regression is a ... Linear Regression Model Assumptions Model Evaluation Metrics Interpretation of Model Coefficients

Work with your assigned Week 4 dataset. It's impossible in various values.

1 Dataset & Setup

Open Colored Dataset assigned to Week 4. Wind the file path.

```
Columns: 9
Variable_1                  <chr> "Life expectancy at birth (years)", "Life expectancy at birth (years)", "Life expectancy at birth (years)", "Life expectancy at birth (years)", "Life expectancy at birth (years)", "Life expectancy at birth (years)", "Life expectancy at birth (years)", "Life expectancy at birth (years)", "Life expectancy at birth (years)", "Life expectancy at birth (years)"...
$ Period                    <dbl> 2021, 2021, 2021,
```

```
(Intercept)  log(GDP_per_capita)
      33.796               4.658
```

▸ Show code

```
Call:
lm(formula = Avg_Life_Expectancy ~ log(GDP_per_capita), data = sixty_Male)

Coefficients:
      (Intercept)  log(GDP_per_capita)
           65.319                1.478
```

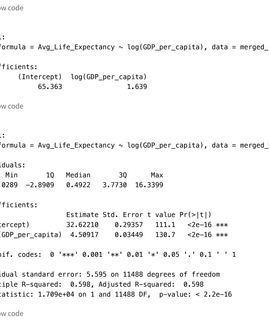
▸ Show code

```
Call:
lm(formula = Avg_Life_Expectancy ~ log(GDP_per_capita), data = sixty_Femal

Coefficients:
      (Intercept)  log(GDP_per_capita)
           65.44                 1.79
```

▸ Show code

▸ Show code



▸ Show code

```
Call:
lm(formula = Avg_Life_Expectancy ~ log(GDP_per_capita), data = merged_birt

Coefficients:
      (Intercept)  log(GDP_per_capita)
           32.622                4.509
```

▸ Show code

```
Call:
lm(formula = Avg_Life_Expectancy ~ log(GDP_per_capita), data = merged_sixt

Coefficients:
      (Intercept)  log(GDP_per_capita)
           65.363                1.639
```

▸ Show code

```
Call:
lm(formula = Avg_Life_Expectancy ~ log(GDP_per_capita), data = merged_birt

Residuals:
     Min       1Q   Median       3Q      Max
-20.0289  -2.8909   0.4922   3.7730  16.3399

Coefficients:
                    Estimate Std. Error t value Pr(>|t|)
(Intercept)           32.6218     0.2935   111.1   <2e-16 ***
log(GDP_per_capita)  4.50917     0.03449   130.7   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 5.595 on 11488 degrees of freedom
Multiple R-squared:  0.598, Adjusted R-squared:  0.598
F-statistic: 1.709e+04 on 1 and 11488 DF,  p-value: < 2.2e-16
```

▸ Show code

```
Call:
lm(formula = Avg_Life_Expectancy ~ log(GDP_per_capita), data = merged_sixt

Residuals:
    Min      1Q  Median      3Q     Max
-9.0317 -1.5080  0.1898  1.5822  8.4054

Coefficients:
                    Estimate Std. Error t value Pr(>|t|)
(Intercept)          65.36291    0.12456  524.8   <2e-16 ***
log(GDP_per_capita)  1.63856    0.01463  112.0   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 2.374 on 11488 degrees of freedom
Multiple R-squared:  0.5218,    Adjusted R-squared:  0.5218
F-statistic: 1.254e+04 on 1 and 11488 DF,  p-value: < 2.2e-16
```

Interpretation (3–4 lines):

- Intercept (β_0) = 32.622 means a person living in a country with 1 USD GDP has a life expectancy of 32 years, the intercept for global life expectancy at 60 is 65.363 years, this means a 60 year old person living in a country with 1 USD GDP is expected to live 5.36 more years.

- Slope (β_1) = 4.509 (for the birth model) means that a 1% increase in a country's GDP per capita is associated with an average increase of 0.045 years in life expectancy at birth. 1.639 (for the age 60 model) means that a 1% increase in a country's GDP per capita is associated with an average increase of 0.016 years in life expectancy at age 60..

- (Optional) Is the slope statistically significant (p-value)? the p values are both 0.0000000000000022, which is less than 0.05 therefore the p value is very significant.

- (Optional) How much variation does the model explain (R^2)? the model explains 59.8% of the variation in the birth data and 52% of the variation in at 60 data.

> **What is a model?**
>
> A **model** is a simplified, mathematical explanation of how things relate or behave.
> It is **expressed mathematically** (equation/algorithm), **summarises** relationships, **simplifies** what matters, **generalises (within limits)**, and **must be interpretable**.
>
> > A good model is clear, honest, and useful — not a perfect copy of reality, but a way to think better about it.

5 Model Fit (Residuals)

Residuals are **errors** = observed − predicted. They should look like random noise around 0.

▸ Show code

Comment (2–3 lines): Are residuals roughly centred around 0 and randomly scattered? Any patterns or extreme points? What does this suggest about model fit?

The residuals for "Model: At 60" are well-centred around 0 and appear randomly scattered, suggesting a good linear fit and that the assumption of constant variance is largely met. In contrast, the residuals for Model: At Birth show a pattern-less cloud , which suggests a non-constant variance, and there are several extreme points (residuals < -20 and > 10) indicating a poor fit compared to the "At 60" model.

5.1 This suggests that while GDP is a significant predictor, a simple linear model is not a great fit for this data. The relationship isn't a perfect straight line, which is why our R-squared was likely not higher.

6 Group Reflection

Keep answers short and specific (1-2 sentences each).

Question	Reflection
Distributions: What did shapes reveal about variation or consistency?	The "At Birth" histogram is broad and somewhat left-skewed, confirming our high SD and the significant variation from early-life mortality. The "At 60" plot is much narrower, more symmetrical, and fits the normal curve better, which proves our earlier finding that variation in longevity is far more consistent for those who survive to 60.
Model: What does your regression line suggest about the relationship?	The regression line shows a positive and statistically significant relationship. It confirms our core hypothesis that as GDP_per_capita increases, Avg_Life_Expectancy (for both models) also tends to increase.
Learning: What was the main thing you learned about your dataset this week?	We learnt that a simple straight-line model is not a great fit for our data. The residual plot showed a pattern-less cloud , proving the relationship isn't perfectly linear and our model is less accurate for high-income countries.

7 Expected Outputs

Your file should include (at least): - Two distribution plots - One scatterplot with fitted regression line - One residual plot - Regression summary with brief interpretation - Completed reflection table

8 You are free to explore further along modeling (e.g., trying different variable combinations, or trying multiple regression, or different models available in R libraries).

9 You may also explore along p-values and R^2.

week05_bias_fairness_ethics_reproducibility

AUTHOR
Group ID: REPLACE_WITH_GROUP_ID

PUBLISHED
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1 Overview

This week's work focuses on the principles of responsible data science. Unlike previous weeks, the focus is more on critical reflection and we will cover:

a. Identifying potential sources of bias in our dataset and workflow.
b. Define and discuss fairness in the context of our project (Life Expectancy vs. GDP).
c. Reflect on "human-in-the-loop" biases (e.g., confirmation, automation).
d. Apply the FATE (Fairness, Accountability, Transparency, Explainability) framework.
e. Implement practical steps to ensure our project is reproducible.

2 Setup

We will use our cleaned dataset prepared in Week 2, which merges WHO Life Expectancy data with World Bank GDP data. We'll use the "at birth" dataset as it showed more variation in our Week 4 analysis

▶ Show code

```
Rows: 11,490
Columns: 9
$ Indicator           <chr> "Life expectancy at birth (years)", "Life expectan…
$ ParentLocation      <chr> "Africa", "Africa", "Africa", "Africa", "Eastern M…
$ Period              <dbl> 2021, 2021, 2021, 2021, 2021, 2021, 2021, 2021, 20…
$ Sex                 <chr> "Male", "Male", "Both sexes", "Male", "Male", "Bot…
$ Avg_Life_Expectancy <dbl> 48.73, 49.57, 51.48, 51.64, 51.75, 52.31, 53.95, 5…
$ Country_Code        <chr> "LSO", "CAF", "LSO", "SWZ", "SOM", "CAF", "SOM", "…
$ Location            <chr> "Lesotho", "Central African Republic", "Lesotho", …
$ GDP_per_capita      <dbl> 1066.5864, 492.2631, 1066.5864, 3966.2986, 549.114…
$ Income_Group        <chr> "Low income", "Low income", "Low income", "Lower m…
```

▶ Show code

```
Rows: 11,490
Columns: 9
$ Indicator           <chr> "Life expectancy at birth (years)", "Life expectan…
$ ParentLocation      <chr> "Africa", "Africa", "Africa", "Africa", "Eastern M…
$ Period              <dbl> 2021, 2021, 2021, 2021, 2021, 2021, 2021, 2021, 20…
$ Sex                 <chr> "Male", "Male", "Both sexes", "Male", "Male", "Bot…
$ Avg_Life_Expectancy <dbl> 48.73, 49.57, 51.48, 51.64, 51.75, 52.31, 53.95, 5…
$ Country_Code        <chr> "LSO", "CAF", "LSO", "SWZ", "SOM", "CAF", "SOM", "…
$ Location            <chr> "Lesotho", "Central African Republic", "Lesotho", …
$ GDP_per_capita      <dbl> 1066.5864, 492.2631, 1066.5864, 3966.2986, 549.114…
$ Income_Group        <chr> "Low income", "Low income", "Low income", "Lower m…
```

3 Bias in the Data Science Workflow

Reflection: The most relevant bias in our project is Sampling Bias. Our Week 2 and 3 analysis noted that our merged dataset has 720 NA values, primarily from missing GDP data. Our project's purpose is to compare income groups, but if low-income countries are systematically more likely to have missing GDP data, they get excluded from the analysis. This would bias our results by under-representing the very group we are most interested in, skewing our "global" averages and potentially minimizing the true scale of inequality.

We cannot recollect this data, so this is Partly in our control. We will document this clearly in our final report, stating the number of countries excluded from the model due to missing data as a key limitation.

4 Fairness, Representation, and Imbalance

4.1 Fairness in Context

Reflection: In our project's context, fairness is best understood as Representation Fairness. Our goal is to analyze global health inequality. If our dataset or models give unequal visibility or performance to certain groups (e.g., Low income countries or the Africa region), our conclusions are not truly global. Our Week 4 residual plots, which showed the model was less accurate for high-income countries , also point to an Outcome Fairness issue , where our model doesn't "work equally well for all groups".

4.2 Detecting Imbalance

We will check the distribution of our key categorical variable, Income_Group, which we created in Week 2.

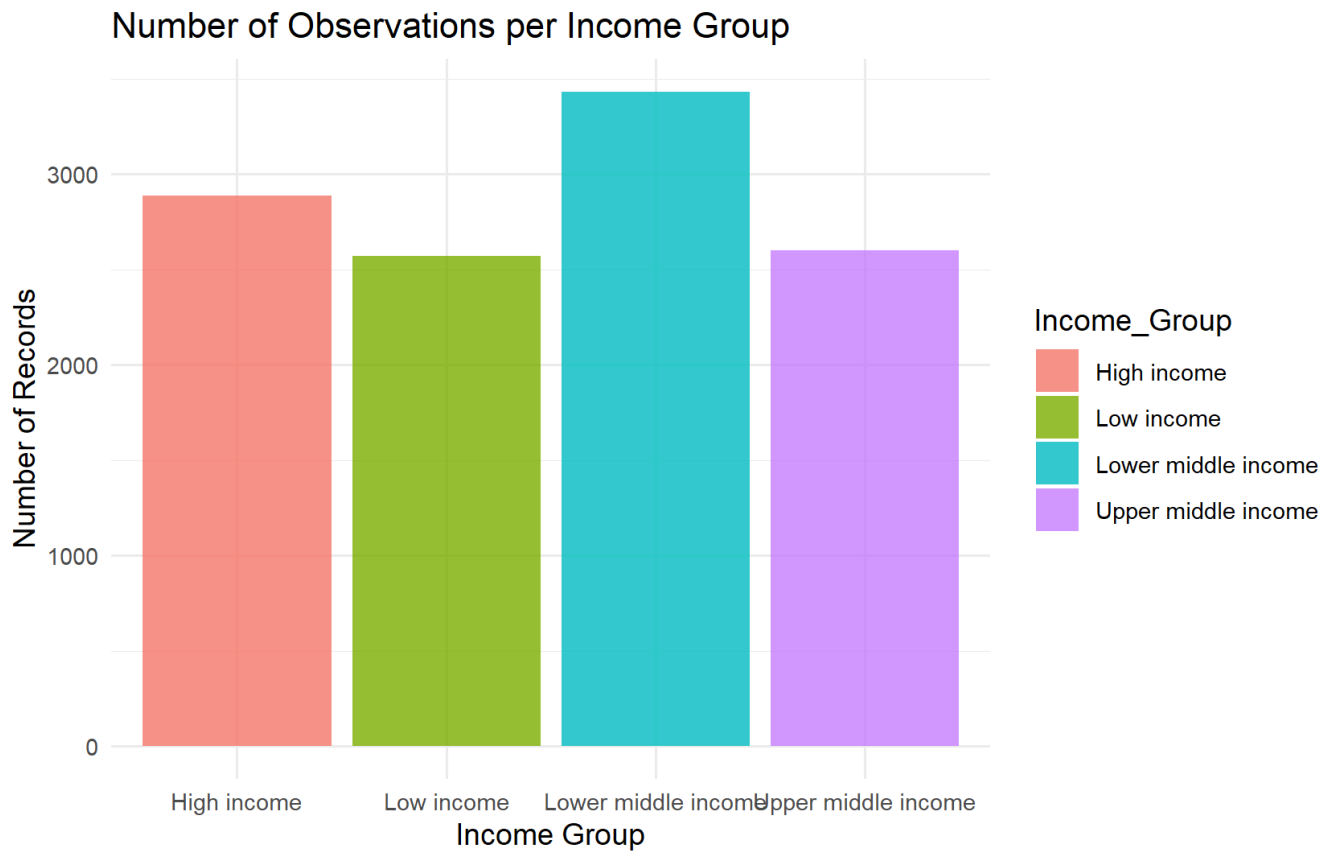
▶ Show code

```
High income          Low income Lower middle income Upper middle income
       2886                2571                3432                2601
```

▶ Show code

```
High income          Low income Lower middle income Upper middle income
       2886                2571                3432                2601
```

▶ Show code



Reflection: Yes, the output shows unequal proportions . The “Lower middle income” group has the most data (3,432 records), while “Low income” has the fewest (2,571 records). This imbalance could create fairness concerns because a global regression model would be more heavily weighted towards the “Lower middle income” group’s data. This could bias our findings by minimizing the impact of the “Low income” group. We will document this limitation and will make sure to report our model’s performance on each income group separately, not just as a single global metric.

5 Human Bias in the Loop

5.1 Confirmation Bias

Reflection: Our main hypothesis, which we tested in Week 4, was: “We expect that log(GDP_per_capita) will show a strong, positive, and statistically significant relationship with Avg_Life_Expectancy at birth”. (Our Week 4 results confirmed this hypothesis).

5.2 Automation Bias

Reflection:We tend to trust our Week 4 model’s p-value < 2.2e-16, which suggests it’s highly significant. However, our own reflection noted the model is unreliable and biased. The residual plot proved the model has a non-constant variance and is less accurate for high-income countries. A human sense-check is to not trust the model’s predictions for wealthy nations and to state that this simple linear relationship breaks down at higher income levels, despite the “significant” p-value.

5.3 Publication Bias

Reflection: In Week 4, we tested both a simple linear model (R^2=0.326) and a log-transformed model (R^2=0.598). We might feel inclined to only share the “better” log-transformed model. However, mentioning that the simple, non-transformed model was a poor fit is crucial. It demonstrates transparency and justifies why the log-transformation was a necessary and appropriate analytical choice, which ultimately increases trust in our final model.

6 Ethical Frameworks in Practice

6.1 FATE Framework

Here is how we have applied the FATE framework in our project:

Principle,Specific Action We Have Taken / Will Take Fairness, We demonstrated Fairness by disaggregating our analysis, checking model performance and distributions for subgroups like Income_Group and ParentLocation.

Accountability, We demonstrate Accountability through our Git commit history and by including our Group ID on all reports. We also state who does what in all our reflective reports and all analysis steps are logged in our weekly .qmd files.”

Transparency, We demonstrated Transparency in our Week 3 report by fully documenting our data sources (WHO, World Bank), their exact licenses (CC BY-NC-SA, CC BY), and our access dates.

Explainability, We demonstrated Explainability in Week 4 by interpreting our log-linear regression slope (1.639) in plain English, not just reporting numbers (e.g., “a 1% increase in…GDP…is associated with an average increase of 0.016 years in life expectancy at age 60”)

6.2 Ethical Domains

Reflection: The ethical domain of Representation connects most directly to our project. As we noted in our Week 3 Risks Reflection, the WHO dataset uses the variable “Sex” (Male, Female, Both sexes), not “Gender”. This misrepresents or excludes populations that do not identify within this binary.

Our concrete action is to be precise in our final report: we have renamed the column Dim1 to Sex (not Gender) and will add a note in our final report clarifying that our analysis is limited to biological sex as defined by the source data, and cannot make claims about gender identity.

7 Reproducibility in Practice

We will now generate a reproducibility log. This log captures our exact software environment, ensuring that anyone (including our future selves) can re-run this analysis under the same conditions.

▶ Show code

```
Log file created at: outputs/week5_session-info.txt
```

▶ Show code

```
[1] TRUE
```

Question	Reflection
Implementation: What reproducibility steps have we completed?	We have used relative paths (e.g. data/processed/) to kept raw data separate, fully documented our data sources in Week 3 and have now generated a week5_session-info.txt file logging the packages
Next Step: What one additional action could we take this week to improve reproducibility?	Our Git commit messages are sometimes vague. We will adopt a stricter convention for commit messages (e.g. ” Add Week 5 FATE analysis”) to make our decision history clearer if anyone needs to revisit a certain version of the work
Audit Connection: What evidence proves our analysis is traceable?	Our audit evidence is our Week 3 Provenance table where our clear file structure is shown (data/raw, data/processed, outputs) and the week5_session-info.txt file.This trail allows an auditor to trace data from its source to our final output.
Long-Term Habit: How will we apply these habits in the future?	We will make a README.md for data provenance and a sessionInfo() log a standard part of all future projects.This is for professional standard for building work that can be trusted and verified by other users.

Review the assignment due date (https://classroom.github.com/a/DXThoYdf) ## `README.md` *(repo guide for students)*

Overview of Data Science Practice – Group Project

Welcome to your group's repository for the **MA7441 Overview of Data Science Practice** module.

This repository will capture your **weekly progress**, **data**, **documentation**, and **outputs**.
It is based on a template maintained at leicester-ods-2025/ods-2025-template (https://github.com/leicester-ods-2025/ods-2025-template).

Group 03

Members:

1. Imraan Morolong
2. Theo Keetile
3. Zubair Mosimanegape Jalal

This repository captures our weekly progress, data, documentation, and outputs as we investigate the relationship between global health outcomes and economic status. ## Project Overview & Rationale:

Global Health Inequality

This project analyzes the relationship between a nation's economic status and its public health outcomes. Our primary research goal is to observe and quantify the differences in life expectancy between high- and low-income countries.

By merging global health data with economic indicators, we aim to: 1. Identify and visualize regional and economic inequalities in life expectancy. 2. Statistically model the strength of the association between GDP and longevity. 3. Explore the potential reasons why these significant gaps in health outcomes exist.

Data Sources

This project integrates two primary datasets, which are stored in data/raw/ and documented in our Week 3 report .

WHO Life Expectancy:

Publisher: World Health Organisation (WHO) Content: Average life expectancy at birth and at age 60, disaggregated by sex, for 185 countries (2000-2021). Accessed: 14 October 2025 License: Creative Commons Attribution-NonCommercial-ShareAlike 3.0 IGO (CC BY-NC-SA 3.0 IGO)

World Bank GDP:

Publisher: World Bank Group (WBG) Content: GDP per capita for over 200 countries (1964-2024). We use this to classify countries into income groups (Low, Lower-middle, Upper-middle, High). Accessed: 15 October 2025 License: Creative Commons Attribution 4.0 (CC BY 4.0)

Key Decisions & Findings Log

This log summarizes our main analytical decisions and findings from each week.

Week 2: Data Wrangling & EDA

Decision: Dropped numerous uninformative columns from the raw WHO data (e.g., Dim2 type, Dim3, FactValueNumericPrefix) that were 100% NA .

Decision: Reshaped the wide-format WBG GDP data into a tidy, long format. This involved pivot_longer() to turn year-based columns (e.g., 1960, 1961…) into key-value pairs (Year, GDP_per_capita), making it mergeable.

Decision: Standardized column names (e.g., SpatialDimValueCode to Country_Code, Period to Year) to create common keys for joining the WHO and WBG datasets.

Decision: Saved the final cleaned and merged datasets to data/processed/ (e.g., merged_table_Birth.csv).

Week 3: Provenance & Risks

Finding (Risk): Identified a key data quality issue. Our merged dataset contains 720 NA values , primarily due to missing GDP data for certain countries/years. This risks sampling bias, as our analysis may unintentionally exclude the low-income countries we are trying to study.

Finding (Ethics): Identified a critical terminology risk. The WHO data provides “Sex” (Male, Female, Both sexes) . Our team had initially (and incorrectly) labeled this variable as “Gender” in Week 2.

Decision (Ethics): We corrected this variable to Sex in our cleaned data and will explicitly state in our final report that our analysis is limited to biological sex as defined by the source, not gender identity.

Finding (License): Identified a licensing constraint. The WHO data’s “Non-Commercial” (CC BY-NC-SA) license applies to our derivative merged dataset, limiting its reuse for commercial purposes.

Week 4: Statistical Modeling

Finding: The relationship between raw GDP_per_capita and Avg_Life_Expectancy is non-linear.

Decision: We log-transformed the predictor variable. The model Avg_Life_Expectancy ~ log(GDP_per_capita) proved to be a much better fit.

Finding: This log-linear model is highly statistically significant p-value < 2.2e-16 and explains a substantial amount of the variance in global life expectancy at birth $R^2 = 0.598$.

Finding: The regression slope for “at birth” data was 4.51 suggesting a strong positive association.

Finding (Limitation): Our residual plots still show a “fan” shape, indicating our model is less accurate for high-income countries and its predictive power is limited.

Week 5: Fairness, Accountability, Transparency, Explainability(FATE) & Reproducibility

Decision: Applied the FATE framework to audit our project’s ethical and responsible practices.

Decision: We confirmed our analysis is reproducible by generating a week5_session-info.txt file. This log contains our R version, OS, and all package versions, allowing an external auditor to replicate our environment precisely.

Challenges and Errors Encountered

This section documents key challenges, errors, and design decisions made during the analysis.

Data Integration:

A major challenge was merging the WHO Life Expectancy data with the World Bank GDP data.

Issue:

The two datasets used different naming conventions for countries and join keys (e.g., `Period` vs. `Year` ,SpatialDimValueCode `vs.` Country Code`).

Solution:

We had to manually rename columns and convert data types (e.g., `Country_Code` to `character`) in both dataframes to ensure a successful `left_join\` .

Handling Non-Country Aggregates:

Issue:

The World Bank GDP data (`GDP_DATA1.csv`) contains many rows that are not countries but regional or financial aggregates (e.g., 'World', 'Euro area', 'Low & middle income').

Solution:

Our initial `left_join` from the life expectancy data to the GDP data automatically filtered out most of these, as they had no matching `Country_Code` in the WHO dataset. For the dynamic income classification, we had to create a specific list (`non_country_groups`) to filter them out explicitly.

Missing Values (NAs):

Issue:

After merging, we found a number of `NA` values for `GDP_per_capita` and `Income_Group` (approx. 5.8% of the data). This was likely due to countries in the WHO data not having corresponding GDP data in the World Bank file for those specific years.

Solution:

For the initial EDA, we chose to remove these rows using `subset()` (or `drop_na()`) to simplify the plots. We acknowledge this is a limitation, and a more advanced approach (like imputation) might be needed for formal modeling.

Misrepresentation of `Dim1` (Sex/Gender):

Issue:

In our initial cleaning (`week2_analysis.qmd`), we renamed the `Dim1` column from the WHO dataset to `Gender` for clarity.

Problem:

As noted in our Week 3 reflection, this was a misrepresentation. The WHO data provides options for "Male," "Female," and "Both sexes," which represent biological **sex**, not the broader social construct of **gender**.

Solution:

We corrected this in our scripts to rename `Dim1` to `Sex`. We acknowledge this is a limitation of the source data, as it means our analysis does not include data on different gender identities.

Critical Data Duplication Error (Week 2):

Issue:

During statistical analysis in Week 4, we discovered that our two final datasets (`Merged_Birth.csv` and `Merged_Sixty.csv`) had identical variance and other descriptive statistics. This was a major red flag.

Root Cause: We traced this back to our `week2_analysis.qmd` script. When creating the final tables, a copy-paste error occurred, and we used the `merged_table_Birth2` dataframe as the source for *both* the `merged_table_Birth` and `merged_table_at_sixty` dataframes. This meant our `Merged_Sixty.csv` file was just an identical copy of `Merged_Birth.csv`.

Contributing Factors: This error was difficult to track due to our group's workflow. Inconsistent R server access forced us to work locally and share code via email, leading to version control issues. Furthermore, our initial exploratory analysis was performed on intermediate filtered dataframes, not on the final saved `.csv` files, so the error was not caught.

Solution:

The error was corrected in the Week 2 script (changing the `subset()` function to use the correct `merged_table_at_sixty2` dataframe). We also had to create an additional income group flag for the 'merged_at_sixty2' data and adjust the naming changes. All further analysis in Week 4 and beyond now loads directly from the corrected `data/processed/` files to ensure all computations are based on the correct, final data. This experience highlighted the importance of robust version control and running validation checks on final outputs.

Folder Structure

Folder / File	Purpose
`_metadata.yml`	Common YAML settings used by all QMD files
`index.qmd`	Main project hub (links to weekly outputs)
`week2_analysis.qmd`	Week 2 scaffold – initial data wrangling
`week3_reflection.qmd`	Week 3 scaffold – reflecting on data from week 2's analysis
`week04_statistics_modelling.qmd`	Week 4 scaffold - statistical Modeling and distribution analysis
`week05_bias_fairness_ethics_reproducibility.qmd`	Week 5 scaffold - bias/fairness reflection, and reproducibility logs
`data/raw/`	Original datasets (read-only)
`data/processed/`	Cleaned datasets saved after wrangling
`docs/`	Additional documentation, notes, screenshots
`outputs/`	Rendered HTML reports
`.gitignore`	Excludes irrelevant files from version control
`.README.md`	A comprehensive guide to the project

Weekly Workflow

1. Work inside your week's `.qmd` file.
2. Save any supporting datasets or figures in the appropriate subfolders.
3. Render your report to `outputs/weekN_report.html`.
4. Push commits with clear messages (e.g., `week2: completed EDA`).

Updating from Template (Upstream)

If your instructor releases updates to the template (e.g., new week files):

```bash git fetch upstream git merge upstream/main
```